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In Defense of Geomancy: Šaraf al-Dīn Yazdī Rebuts Ibn Ḫaldūn's Critique of the Occult Sciences

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Abstract

The late 8th/14th century saw a renaissance of high occultism throughout Islamdom—a development alarming to puritan scholars. This includes Ibn Ḫaldūn (d. 808/1406), whose anti-occultist position in the *Muqaddima* is often assumed to be an example of his visionary empiricism; yet his goal is simply the recategorization of all occult sciences under the twin rubrics of magic and divination, and his veto persuades more on religious and social grounds than natural-scientific. Restoring the historian's argument to its original state of debate with the burgeoning occultist movement associated with the Mamluk sultan Barqūq's (r. 784/1382-791/1389 and 792/1390-801/1399) court reveals it to be not forward-thinking but rather conservative, fideist and indeed reactionary, as such closely allied with Ibn Qayyim al-Ǧawziyya's (d. 751/1350) puritanical project in particular; and in any event, the eager patronage and pursuit of the occult sciences by early modern ruling and scholarly elites suggests that his appeal could only fall on deaf ears. That it also flatly opposed the forms of millennial sovereignty that would define the post-Mongol era was equally disqualifying. I here take Šaraf al-Dīn ʿAlī Yazdī (d. 858/1454), Ibn Ḫaldūn's younger colleague and fellow resident in Cairo, as his sparring partner from the opposing camp: the Timurid historian was a card-carrying occultist and member of the Iḫwān al-Ṣafāʾ network of neopythagorean-neoplatonic-monist thinkers then gaining prominence from India to Anatolia via Egypt. I further take geomancy (*ʿilm al-raml*) as a test case, since Yazdī wrote a tract in defense of the popular divinatory

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science that directly rebuts Ibn Ḫaldūn's arguments in the *Muqaddima*. To set the stage for their debate, I briefly introduce contemporary geomantic theory and practice, then discuss Ibn Ḫaldūn's and Yazdī's respective theories of occultism with a view toward establishing points of agreement and disagreement; I also append a translation of Yazdī's tract as a basis for this comparison.

Keywords

Ibn Ḫaldūn, *Muqaddima*, Šaraf al-Dīn ʿAlī Yazdī, Sulṭān Barqūq, Mamluk Cairo, occult sciences (*al-ʿulūm al-ġarība*), geomancy (*ʿilm al-raml*), lettrism (*ʿilm al-ḥurūf*), magic, divination, neopythagoreanism, neoplatonism, monism, theories of history, millennial sovereignty

Résumé

La fin du VIIIe/XIVe siècle vit une renaissance de l'occultisme d'élite en terre d'Islam, un développement alarmant aux yeux des savants puritains, y compris Ibn Ḫaldūn (m. 808/1406), dont les positions anti-occultistes dans sa *Muqaddima* sont souvent considérées comme un exemple de son empirisme visionnaire. Pourtant, son but est simplement de reclasser toutes les sciences occultes sous les étiquettes gémellaires de magie et de divination. Son opposition catégorique convainc davantage sur des bases religieuses et sociales que naturelles et philosophiques. Resituer l'argumentaire d'Ibn Ḫaldūn dans son débat d'origine avec le mouvement occultiste émergent associé à la cour du sultan mamelouk Barqūq (r. 784/1382-791/1389 et 792/1390-801/1399) révèle qu'il ne s'agit pas d'une réflexion d'avant-garde mais plutôt conservatrice, dogmatique et, en fait, réactionnaire, puisque particulièrement en étroite relation avec le projet puritain d'Ibn Qayyim al-Ǧawziyya (m. 751/1350) en particulier. Dans tous les cas, le mécénat actif et la quête des sciences occultes par les élites dirigeantes et savantes du début de l'époque moderne suggèrent que son appel ne pouvait que tomber dans l'oreille de sourds. En outre, le fait qu'il s'oppose catégoriquement aux formes de souveraineté millénaire qui définissaient l'époque post-mongole était également disqualifiant. Je prends ici Šaraf al-Dīn ʿAlī Yazdī (m. 858/1454), le jeune collègue d'Ibn Ḫaldūn résidant au Caire, en tant qu'adversaire : l'historien timouride était un occultiste convaincu et membre du réseau de penseurs néopythagoriciens, néoplatoniciens et monistes des Iḫwān al-Ṣafāʾ qui gagnaient alors en importance de l'Inde à l'Anatolie en passant par l'Égypte. Je prendrai la géomancie (*ʿilm al-raml*) comme un cas d'école, puisque Yazdī écrivit un opuscule pour défendre la science divinatoire populaire qui réfute directement les arguments

d'Ibn Ḫaldūn dans sa *Muqaddima*. Pour poser le débat, je présenterai succinctement la théorie et la pratique géomantiques contemporaines, puis discuterai respectivement des théories sur l'occultisme d'Ibn Ḫaldūn et de Yazdī en vue d'établir des points d'accord et de désaccord. J'ajouterai également à cette comparaison une traduction de l'opuscule de Yazdī.

Mots clefs

Ibn Ḫaldūn, *Muqaddima*, Šaraf al-Dīn ʿAlī Yazdī, Sulṭān Barqūq, Le Caire mamelouk, sciences occultes (*al-ʿulūm al-ġarība*), géomancie (*ʿilm al-raml*), science des lettres (*ʿilm al-ḥurūf*), magie, divination, néopythagorisme, néoplatonisme, monisme, théories de l'histoire, souveraineté millénaire

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> The world is full of matters occult.
>
> IBN ḪALDŪN[1]

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The scene is 1390s Mamluk Cairo; our principals are two historians living in that magnetic and stimulating city, the elder hailing from Tunis and the younger from Yazd, who advance what seem on the face of it to be diametrically opposed views on the theory and practice of the occult sciences.[2] The first is Ibn

1 Ibn Ḫaldūn, *Muqaddima*, ed. ʿAbd al-Salām al-Šaddādī, Casablanca, Bayt al-funūn wa-l-ʿulūm wa-l-ādāb, 2005, I, p. 168. This translates Ibn Ḫaldūn's frank if tendentious admission in his discussion of various techniques of psychic perception: *wa-l-ʿālam abū l-ġarāʾib*. Franz Rosenthal's translation of *ġarāʾib* here as "remarkable things" understates the matter, given the specifically occult (*ġarīb*) context in which the phrase occurs. Ibn Ḫaldūn, *The Muqaddimah: An Introduction to History*, transl. Franz Rosenthal, Princeton, Princeton University Press, 1967, I, p. 217.

2 The standard Arabic term for the occult sciences, including astrology (*aḥkām al-nuǧūm*), alchemy (*kīmiyāʾ*) and a variety of magical and divinatory techniques, is *ʿulūm ġarība*, meaning those sciences that are unusual, rare or difficult, *i.e.* elite; less frequently used terms are *ʿulūm ḫafiyya* and *ʿulūm ġāmiḍa*, sciences that are hidden or occult. These terms are routinely used in classifications of the sciences, biographical dictionaries, chronicles, etc. Its 19th-century European flavor notwithstanding, the term occultism is therefore used here simply to

Ḫaldūn (d. 808/1406), feted theoretician of history and occasional statesman, and the second Šaraf al-Dīn ʿAlī Yazdī (d. 858/1454), dynastic historian to the Timurids and committed occultist.[3]

Ibn Ḫaldūn's purported objectivity and skepticism on the subject of the occult sciences is now much admired, even held up as a shining, solitary example of rationality amidst rampant superstition; a man ahead of his time, he famously rejects well-patronized sciences like astrology and alchemy out of hand.[4] What is less acknowledged, however, is the fact that the Tunisian historian argues forcefully, and at length, for the incontrovertible reality of magic and divination with the selfsame sober objectivity.[5] For their part, Yazdī's theories of historical evolution and human potential are not dissimilar to Ibn Ḫaldūn's more famous formulations, yet are predicated precisely on a thoroughgoingly occultist outlook. Both men's outspoken views on the subject of occultism, in turn, go to the heart of their respective sociologies and political theologies, which accordingly converge and diverge in instructive ways. Restoring Ibn

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denote a scholarly preoccupation with one or more of the occult sciences as discrete natural-scientific or mathematical disciplines.

3 Yazdī is best known as the author of the *Ẓafarnāma*, an ornate, long-admired and much-imitated account of Tīmūr's career; it is less widely known that this is one of five separate chronicles that he began, including a *Muqaddima, Fatḥnāma-yi ṣāḥib-qirānī, Fatḥnāma-yi humāyūn* and the *Second Maqāla*, none of which he finished and whose shifting ideological strategies reflect the displacement of Tīmūr's dispensation by that of his son Šāhruḫ. See İ. Evrim Binbaş, *Intellectual Networks in Timurid Iran: Sharaf al-Dīn ʿAlī Yazdī and the Islamicate Republic of Letters*, Cambridge, Cambridge University Press ("Cambridge studies in Islamic civilization"), 2016, p. 199-250.

4 This most recently and explicitly in Ahmad Dallal's *Islam, Science, and the Challenge of History*, New Haven-London, Yale University Press ("The Terry lectures"), 2010, which summary of the state of the field of history of Islamicate science mentions ubiquitous sciences like alchemy and magic but once—as being forever exiled to the shadowlands of unscience by Ibn Ḫaldūn (p. 144-145). Such an argument, of course, is formally identical to the long-outmoded claim that al-Ġazālī (d. 505/1111) was responsible for the permanent crippling of philosophy within Islamdom, and merely testifies to the positivist occultophobia still dominating the field.

5 As Robert Irwin remarks, "It is easy to exaggerate the modernity of Ibn Khaldūn by discounting his flirtations with the supernatural, as well as his intense, though rather conventional piety". Robert Irwin, "Al-Maqrīzī and Ibn Khaldūn, Historians of the Unseen," *Mamlūk Studies Review*, 7/2 (2003), p. 222. Ḥasan Qubaysī offers a critique of this tendentious narrative in his *al-Matn wa-l-hāmiš: Tamārīn ʿalā l-kitāba al-nāsūtiyya*, Casablanca, al-Markaz al-ṯaqāfī l-ʿarabī, 1997. On the various contemporary, early modern Ottoman and modern European and Arab receptions of Ibn Ḫaldūn see Róbert Simon, *Ibn Khaldūn: History as Science and the Patrimonial Empire*, transl. K. Pogátsa, Budapest, Akadémiai Kiadó ("Bibliotheca Orientalis Hungarica", 48), 2002, p. 11-80.

Ḫaldūn's and Yazdī's writings to their original state of dialogue—again, the two historians possibly met in Cairo, and certainly moved in the same court circles while there, especially that of al-Malik al-Ẓāhir Barqūq (r. 784/1382-791/1389 and 792/1390-801/1399)[6]—thus furnishes us with several hitherto unexploited advantages. Most significantly, we stand to gain more traction on Ibn Ḫaldūn's larger project by understanding that to which he was *reacting*—or as Yazdī would say, *overreacting*—in his anti-occultism jags, among the most neglected elements of the *Muqaddima*.[7]

For reasons of space, I approach this problematic here by taking geomancy (*ʿilm al-raml*) as a convenient case study, since Ibn Ḫaldūn and Yazdī each spend a few pages discussing this immensely popular divinatory art in ways that debouch revealingly into the wider field here in view. To this end, I first define this occult science and briefly sketch the contours of the Islamicate geomantic tradition, with a focus on the consolidating Persianate branch of that tradition during the 7th/13th-10th/16th centuries. I then discuss Ibn Ḫaldūn's and Yazdī's respective theories of occultism with a view toward establishing points of philosophical agreement and disagreement. Finally, I translate a tract Yazdī wrote in defense of geomancy that gives the distinct impression of having been addressed in the first instance to Ibn Ḫaldūn.

6 Ibn Ḫaldūn lived in Cairo from 785/1383 to his death in 808/1406, and Yazdī spent at least part of the period 795/1393-810/1408 there with his teacher and friend Ibn Turka (d. 835/1432). See Matthew Melvin-Koushki, *The Quest for a Universal Science: The Occult Philosophy of Ṣāʾin al-Dīn Turka Iṣfahānī (1369-1432) and Intellectual Millenarianism in Early Timurid Iran*, PhD dissertation, Yale University, 2012, p. 34, 46-52; Walter J. Fischel, *Ibn Khaldūn in Egypt: His Public Functions and His Historical Research (1382-1406): A Study in Islamic Historiography*, Berkeley-Los Angeles, University of California Press, 1967.

7 Several scholars have discussed this aspect of Ibn Ḫaldūn's project, though with insufficient attention to contemporary high occultist theory; as Asatryan's study is the most thorough of these, I depend on it in the main in my discussion of Ibn Ḫaldūn's theory of occultism below. See Mushegh Asatrian (Asatryan), "Ibn Khaldūn on Magic and the Occult," *Iran and the Caucasus*, 7/1-2 (2003), p. 73-123; Robert Irwin, "Al-Maqrīzī and Ibn Khaldūn"; Yves Marquet, "Religion, philosophie et magie devant Ibn Ḫaldūn," *Studia Islamica*, 62 (1985), p. 145-153; *id.*, "Ibn Ḫaldūn et les conjonctions de Saturne et de Jupiter," *Studia Islamica*, 65 (1987), p. 91-96; Aziz Al-Azmeh, *Ibn Khaldūn: An Essay in Reinterpretation*, London, Cass, 1982; Toufic Fahd, *La divination arabe: études religieuses, sociologiques et folkloriques sur le milieu natif de l'Islam*, Leiden, Brill, 1966, p. 45-50, 196-204; and Abderrahmane Lakhsassi, "Magie: le point de vue d'Ibn Khaldûn," in *Coran et talismans: Textes et pratiques magiques en milieu musulman*, ed. Constant Hamès, Paris, Karthala, 2007, p. 95-112.

Islamicate Geomancy[8]

Geomancy, the 'science of sand' (*ʿilm al-raml*), is a uniquely Arabic and Islamicate occult science that captured the intellects and imaginations of scholarly elites and their royal patrons throughout the premodern Islamo-Christianate world. As a complex divinatory science based on a binary code, geomancy stands precise cognate to the ancient Chinese *I Ching*, but commanded in its prime a far greater territorial spread: it was—and in many cases still is—regularly practiced as a single tradition from Fez, Paris and Timbuktu to Kashgar, Kabul and Delhi, and calved simpler versions throughout sub-Saharan Africa and thence the western hemisphere that remain very much in use. Throughout its Afro-Eurasian domain geomancy was third in popularity only to astrology and oneiromancy; indeed, due to its early incorporation of astrological correspondences, it was often considered a form of "terrestrial astrology" requiring much less astronomical expertise while remaining richly informative.[9] This association is exemplified by the fact that in the early modern Persianate East individuals designated *munaǧǧim*s, or court astrologers, increasingly acted as court geomancers as well.[10] *Raml* and *ǧafr*, or letter divination, were also often seen as a complementary pair of divinatory techniques based on number theory (*ʿilm al-ʿadad*),[11] the first employing dots (*niqāṭ*) and figures (*aškāl*) and the second the letters (*ḥurūf*) of the Arabic alphabet (also derived from a dot), and both equally productive of knowledge of past, present and future.

8 This section is largely taken from my study "Persianate Geomancy from Ṭūsī to the Millennium: A Preliminary Survey," forthcoming in *Occult Sciences in Premodern Islamic Culture*, ed. Nader El-Bizri and Eva Orthmann, Beirut, Orient-Institut Beirut, 2017; interested readers will find a fuller treatment of the subject there.

9 See *e.g.* Ibn Ḫaldūn, *Muqaddima*, I, p. 174-179. Significantly, in his classification of the sciences, Ḥāǧǧī Ḫalīfa categorizes geomancy under the rubric of judicial astrology. Fahd, *La divination arabe*, p. 41.

10 *Ibid.*, p. 200. This blending of geomancy and astrology began early on; one of the first codifiers of the Arabic geomantic tradition, Ibn Maḥfūf (d. before 664/1265), is typically called al-Munaǧǧim. *Ibid.*, p. 201.

11 That is, the "science of number," or arithmetic, particularly as associated with Pythagoras and Nicomachus. The 4th/10th-century group of neopythagorean authors known as the Brethren of Purity, for instance, vigorously assert arithmetic to be the basis for all other mathematical, natural, psychological and metaphysical sciences, an ascending epistemological series that culminates in sciences like astrology, alchemy and magic; see Nader El-Bizri, "Epistolary Prolegomena: On Arithmetic and Geometry," in *Epistles of the Brethren of Purity: The Ikhwān al-Ṣafāʾ and Their Rasāʾil: An Introduction*, ed. Nader El-Bizri, Oxford, Oxford University Press, 2008, p. 180-213.

Knowledge of geomancy entered the Christianate world in the 6th/12th century with Latin translations of Arabic works, its popularity burgeoning from there.[12] In the Islamicate world, geomancy was typically associated in the first place with the prophets Idrīs (Enoch or Hermes)[13] and Daniel and the Indian sage Ṭumṭum, not to mention a number of other standard occultist authorities.[14] Intellectual genealogies of the science in Arabic and Persian works on the subject thus presuppose a pre-Islamic Near Eastern or Indian origin, as well as an early Berber connection; the otherwise unknown Abū ʿAbd Allāh Muḥammad al-Zanātī (*fl.* before 629/1230), presumably of the Berber Zanāta tribe, is acclaimed as the first major Arabic exponent of geomancy.[15]

While geomancy fell out of mainstream use in post-Enlightenment Europe,[16] it experienced no such decline in the "un-Enlightened" Islamicate

12 The extant Latin treatises are surveyed in Thérèse Charmasson, *Recherches sur une technique divinatoire: la géomancie dans l'Occident médiéval*, Genève-Paris, Droz-H. Champion ("Hautes études médiévales et modernes", 44), 1980. After Hugh of Santalla's (*fl.* 1145) pathbreaking translations of two treatises from the Arabic under the titles *Ars geomantiae* and *Geomantia nova*, notable examples include Heinrich Cornelius Agrippa's (d. 1535) *In Geomanticam disciplinam lectura* (*Opera*, Hildesheim, Olms, 1970, I, p. 500-526), Christophe de Cattan's (*fl.* 1558) *La Géomance … avec la roüe de Pythagoras* (Paris, 1558; English translation by Francis Sparry in 1591) and Robert Fludd's (d. 1637) *De geomantia* (Veronae, 1687, p. 1-170). See Marion B. Smith, "The Nature of Islamic Geomancy with a Critique of a Structuralist's Approach," *Studia Islamica*, 49 (1979), p. 5-38: p. 8; Stephen Skinner, *Geomancy in Theory and Practice*, St. Paul, Golden Hoard Press, 2011, p. 112-116.

13 In his *Nafaḥāt al-asrār*, a comprehensive geomantic manual written in Damascus in 1277/1861, Muḥammad Bāqir Ṭabāṭabāʾī Yazdī invokes a statement by Imam ʿAlī attributing the science to Idrīs as received from Gabriel. Muḥammad Bāqir Ṭabāṭabāʾī Yazdī, *Nafaḥāt al-asrār*, lithograph, Najaf, Dār al-kutub al-ʿirāqiyya, 1359/1940, p. 10.

14 These include Imams ʿAlī b. Abī Ṭālib and Ǧaʿfar al-Ṣādiq as well as a number of other biblical prophets. On Ṭumṭum al-Hindī see Anton Hauber, "Ṭomṭom (Ṭimṭim) = *Dandamiz = Dindymus*?," *Zeitschrift der Deutschen Morgenländischen Gesellschaft*, 63 (1909), p. 457-472.

15 The best study to date of the Arabic geomantic tradition through the 7th/13th century is Emilie Savage-Smith and Marion B. Smith, "Islamic Geomancy and a Thirteenth-Century Divinatory Device: Another Look," in *Magic and Divination in Early Islam*, ed. Emilie Savage-Smith, Aldershot, Ashgate, 2004, p. 211-276, which expands upon their findings in a similarly titled publication from 1980; in particular, they show that it was fully developed and established by the middle of that century.

16 Geomancy was largely replaced by the *I Ching* as the divinatory art of choice in modern Euro-American occultism, particularly as represented by the Hermetic Order of the Golden Dawn; this Chinese cognate was also the subject of considerable interest to C.G. Jung (along with a number of other occult sciences), who postulated that number regulates both psyche and matter and described divination as the "science of synchronicity." See

world, and particularly its vast Persianate subset, where occultist traditions enjoyed smoother continuity and wider practice. Along with astrologers and experts in *ǧafr*, geomancers (sg. *rammāl*) were in high demand at imperial and regional courts during the early modern period—witness Ḥaydar Rammāl's successful and influential career at the court of Sulṭān Süleymān (r. 926/1520-974/1566) in Anatolia,[17] that of Hidāyat Allāh Munaǧǧim-i Šīrāzī at the court of Emperor Akbar (r. 963/1556-1014/1605) in India, or that of Ǧalāl al-Dīn Munaǧǧim-i Yazdī, author of the *Tārīḫ-i ʿAbbāsī*, at the court of Šāh ʿAbbās (r. 995/1587-1038/1629) in Iran.[18] Well into the modern period we find vigorous testimony that the science was still considered a scholarly staple from the Maghrib to India; in 13th/19th-century Samarkand and Bukhara, for instance, surviving miscellany notebooks kept by judges suggest that they often employed geomantic readings to help them decide court cases.[19] Geomancy's popularity remains unabated in Iran today, though frequent abuse by fraudsters has made its practice into something of a social problem; as a result, *rammālī* ('geomancing') is now legally punishable by a fine and up to seven years' imprisonment.[20] In its current, flabby Persian usage *rammālī* usually connotes witchcraft or hocus-pocus, particularly of the hornswoggle

e.g. Marie-Luise von Franz, *On Divination and Synchronicity: The Psychology of Meaningful Chance*, Toronto, Inner City Books ("Studies in Jungian psychology", 3), 1980. More recently, however, geomancy appears to be experiencing a modest revival in the Anglo-American world; witness John Michael Greer's *The Art and Practice of Geomancy: Divination, Magic, and Earth Wisdom of the Renaissance*, San Francisco, Weiser, 2009, a comprehensive and highly usable manual in the early modern occult-philosophical tradition.

17 See Cornell Fleischer, "Shadow of Shadows: Prophecy in Politics in 1530s Istanbul," *International Journal of Turkish Studies*, 13 (2007), p. 51-62; *id.*, "Seer to the Sultan: Haydar-i Remmal and Sultan Süleyman," in *Cultural Horizons: A Festschrift in Honor of Talat S. Halman*, ed. Jayne L. Warner, Istanbul-Syracuse, Syracuse University Press, 2001, I, p. 290-299.

18 On the latter two geomancers see Melvin-Koushki, "Persianate Geomancy."

19 They also appear to have regularly used lettrist techniques to analyze people's character on the basis of their names. More generally, biographical dictionaries for 13th/19th-century Transoxania indicate that some five percent of scholars were known to be practicing occultists—the same percentage that are identified as physicians. My thanks to James Pickett for bringing this evidence to my attention; see now our "Mobilizing Magic: Occultism in Central Asia and the Continuity of High Persianate Culture under Russian Rule," *Studia Islamica*, 111/2 (2016), p. 231-284.

20 Alireza M. Doostdar, *Fantasies of Reason: Science, Superstition, and the Supernatural in Iran*, PhD dissertation, Harvard University, 2012, p. 48 n. 19; see also his forthcoming *The Iranian Metaphysicals: Explorations in Science, Islam, and the Uncanny*, Princeton, Princeton University Press, 2017.

variety, though is not as wholly negative as the term *ǧādūgarī*.[21] In 2009, for instance, opposition candidate Mir Hossein Mousavi derided Mahmoud Ahmadinejad's government as one of *kaf-bīnī u rammālī*, or palmistry and poppycock.[22] Yet even such infamy is an index of the continued inextricability of geomancy from Persianate culture.

The various Arabic terms for geomancy, the 'science of sand' (*ʿilm al-raml, ḫaṭṭ al-raml, ḍarb, ṭarq*),[23] all refer to its basic procedure of drawing 16 random series of lines in the sand or dirt to generate the first four tetragrams of a geomantic reading. Confusingly, in modern English usage, geomancy can also refer to the Chinese art of *feng shui*, though this is a misnomer; as a system of divining the subtle currents of the earth for the purposes of building or burying, it is more accurately termed 'topomancy.'[24] As with *I Ching* trigrams, the four lines of a geomantic figure (*šakl*) are generated by the odd (*fard*) or even (*zawǧ*) result of each line, creating a binary code represented as either one dot (*nuqṭa*) or two dots respectively—hence the science's alternative name of *ʿilm al-nuqṭa* or *ʿilm al-niqāṭ*.[25] This binary code is then deployed according to set procedures to capture the flux patterns of the four elemental energies (fire, air, water, earth) as a means to divine past, present and future events. Emilie Savage-Smith summarizes the geomantic method as follows:

> The divination is accomplished by forming and then interpreting a design, called a geomantic tableau, consisting of 16 positions, each of which is occupied by a geomantic figure. The figures occupying the first four positions are determined by marking 16 horizontal lines of dots on a piece of paper or a dust board. Each row of dots is examined to determine if it is odd or even and is then represented by one or two dots accordingly. Each figure is then formed of a vertical column of four marks, each of which is either one or two dots. The first four figures, generated by lines made while the questioner concentrates upon the question, are

21 Doostdar, *Fantasies*, p. 46 n. 17.

22 *Ibid.*, p. 4.

23 Toufic Fahd, "Khaṭṭ," *EI*². *ʿIlm al-raml* is occasionally translated as *ʿilm-i rīg* in Persian.

24 Skinner, *Geomancy*, p. 36. By analogy with *feng shui*, exponents of ley-line theory also adopted the term geomancy for their practice, further muddying the waters.

25 As Stephen Skinner notes in his manual on the subject, "In this century when computers now make many of man's economic, political and commercial forecasts, it is easy to forget that these machines work on the same principle of binary mathematics as the infinitely more ancient machines of the *I Ching* and geomancy." Skinner, *Geomancy*, p. 35.

FIGURE 1 *A typical geomantic tableau.*[26]

> placed side by side in a row from right to left. From these four figures the remaining 12 positions in the tableau are produced according to set procedures. Various interpretative methods are advocated by geomancers for reading the tableau, often depending upon the nature of the question asked.[27]

From right to left, the first four figures in the top row are termed Mothers (*ummahāt*), which are combined to produce the second four in the same row, termed Daughters (*banāt*); the four figures the Mothers and Daughters produce in the next row are termed Nieces (*ḥafīdāt, mutawallidāt*); and the Nieces are combined to produce the *zawāʾid* in the rows below: first, the Right and Left Witnesses (*šawāhid*), which in turn produce the Judge (*mīzān*) in the 15th position at the bottom of the geomantic tableau or shield chart (*taḫt*). In the 16th and final position, usually drawn below and to the right of the Judge, is the Reconciler (*ʿāqiba*), produced by the combination of the first Mother and the Judge.

The number of possible combinations of figures in a geomantic tableau is 16^4, or 65,536 in all.[28] Each of the 16 geomantic figures and the 15 houses of a tableau acquired specific elemental, astrological, calendrical, numerical, lettrist, humoral, physiognomical and other correspondences; detailed

26 This example is from the margin of Princeton's copy of Šams al-Dīn Ḫafrī's *Risāla dar Raml*, MS New Series 1177, f. 24b. Note that, as here, a line of two dots in a geomantic figure is usually written as a dash.

27 Emilie Savage-Smith, "Geomancy in the Islamic World," in *Encyclopaedia of the History of Science, Technology, and Medicine in Non-Western Cultures*, ed. Helaine Selin, Berlin-Heidelberg-New York, Springer, 2008, p. 998.

28 Smith, "The Nature of Islamic Geomancy," p. 14.

information can thus be derived from the figures about virtually any aspect of human experience, whether physical, mental or spiritual, whether past, present or future.

Although many techniques of divination long persisted in the Islamicate world,[29] *raml*, with *ǧafr* and quranic bibliomancy (*fāl-i Qurʾān*),[30] has the distinction of being the most fully Islamicized; this trio soon outpaced the others in prestige and elite patronage as a result.[31] Geomantic treatises frequently begin by marshaling a number of prooftexts from the Quran and Hadith in defense of the science's religious legitimacy.[32]

The work of Emilie Savage-Smith, Marion B. Smith and Toufic Fahd in particular marks the indispensable starting point for any inquiry into Islamicate geomancy;[33] Robert Jaulin has explored the mathematical properties of the science from a structuralist standpoint,[34] and a number of anthropologists have conducted surveys of geomantic practice in modern-day sub-Saharan Africa and Yemen, including derivative forms such as *ifa, gara* or *sikidy*.[35] What

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29 Scapulomancy (*ʿilm al-katif*), for instance, or divination by shoulder blade, was frequently used in Khʷārizm until the modern period to predict the outcome of caravan journeys, resolve legal cases, etc. (personal communication with James Pickett).

30 Bibliomancy, especially in reference to the Qurʾān or the *dīwān* of Ḥāfiẓ (d. *ca* 792/1390), similarly flourished in the Persianate world during the 7th/13th-10th/16th centuries; see *e.g.* Christiane Gruber, "The 'Restored' Shīʿī *muṣḥaf* as Divine Guide? The Practice of *fāl-i Qurʾān* in the Ṣafavid Period," *Journal of Qurʾānic Studies*, 13/2 (2011), p. 29-55.

31 Fahd, *La divination arabe*, p. 204.

32 This includes Kor 46, 4: "Bring me a Book before this, or some trace of a science (*aṯāratⁱⁿ min ʿilm*), if you speak truly." And the following hadith is constantly cited: "There was a prophet [who practiced divination by] drawing lines [in the sand], so one may draw the way he did" (see *e.g.* al-Buḫārī, *Ṣaḥīḥ*, *Kitāb al-Masāǧid wa-mawāḍiʿ al-ṣalāt*, bab 8, nº 1227; Abū Dāwūd, *Sunan*, *Kitāb al-Ṣalāt*, bab 173, nº 931; etc.).

33 See above; other studies include Felix Klein-Franke, "The Geomancy of Aḥmad b. ʿAlī Zunbul: A Study of the Arabic Corpus Hermeticum," *Ambix*, 20/1 (March 1973), p. 26-35; Valerio Cappozzo, "Libri dei sogni e geomanzia: la loro applicazione letteraria tra Islam, medioevo romanzo e Dante," *Quaderni di Studi Indo-Mediterranei*, 2 (2009), p. 207-226.

34 Robert Jaulin, *La géomancie: analyse formelle*, Paris, Mouton, 1966; *id.*, *Géomancie et Islam*, Paris, Bourgois, 1991. Smith's "The Nature of Islamic Geomancy" is a critique of Jaulin's studies.

35 See *e.g.* Bernard Maupoil, "Contribution à l'étude de l'origine musulmane de la géomancie dans le Bas-Dahomey," *Journal de la Société des Africanistes*, 13 (1943), p. 1-94; Jacques Faublée, "Techniques divinatoires et magiques chez les Bara de Madagascar," *Journal de la Société des Africanistes*, 21 (1951), p. 127-138; Robert Jaulin, "Essai d'analyse formelle d'un procédé géomantique," *Bulletin de l'Institut Français d'Afrique Noire*, series B, 19 (1957), p. 43-71; William R. Bascom, *Ifa Divination: Communication Between Gods and Men in West*

has been almost wholly neglected to date, however, is the *Persianate* geomantic tradition.[36] This vital and productive tradition was the driving force behind the florescence of Islamicate geomancy from the 8th/14th century onward, itself part of the great renaissance of occultism throughout the post-Mongol Islamicate world, and appears to begin in earnest in the mid-7th/13th century with the importation to Iran of the Maghribi geomantic tradition associated in the first place with al-Zanātī. This tradition would remain the gold standard for centuries: the 7th/13th-century Shirazi geomancer Nāṣir b. Ḥaydar cleaves to the method of the Berber master, and even three centuries later the Isfahani-Tabrizi geomancer Ḥaydar b. Muḥammad felt it necessary to mention that his brilliant teacher in the science was from the Maghrib.[37] More famously, Naṣīr al-Dīn al-Ṭūsī (d. 672/1274), the indefatigable philosopher, mathematician

Africa, Bloomington, Indiana University Press, 1969; Philip M. Peek (ed.), *African Divination Systems: Ways of Knowing*, Bloomington, Indiana University Press ("African systems of thought "), 1991; Annick Regourd, "Pratiques de géomancie au Yémen," in *Religion et pratiques de puissance*, ed. Albert de Surgy, Paris, Éditions L'Harmattan, 1997, p. 105-128; Jan Jansen, "Éducation arithmétique sous forme d'apprentissage: La géomancie dans les Monts Mandingues," *Cahiers d'Études africaines*, 51/1, 201 (2011), p. 9-49; see Smith, "The Nature of Islamic Geomancy," p. 9.

36 Apart from the early 20th-century surveys of Persian folklore by Henri Massé and Bess Allen Donaldson which are largely incognizant of more elite, scholarly forms of geomantic practice, the only attention that has been paid to the Persianate geomantic tradition is Charles Ambrose Storey's list of works on the subject but as a strictly preliminary foray it is unsystematic and incomplete, and focuses primarily on Indian collections. Henri Massé, *Persian Beliefs and Customs*, transl. Charles Messner, New Haven, Human Relations Area Files, 1954; on *raml* see p. 249-250; Bess Allen Donaldson, *The Wild Rue: A Study of Muhammadan Magic and Folklore in Iran*, London, Luzac, 1938; on *raml* see p. 194-195; Charles Ambrose Storey, *Persian Literature: A Bio-Bibliographical Survey*, Leiden, Brill, 1977, II/3, p. 480-489. Emilie Savage-Smith and Marion B. Smith have noted the existence of a healthy Persianate geomantic tradition from Faḫr al-Dīn al-Rāzī and Naṣīr al-Dīn al-Ṭūsī onward but have dealt to date primarily with late medieval Arabic geomancy. This general neglect has largely to do with the fact that Persian geomantic texts were not transmitted to Europe, as well as the fact that they are an important component of the renaissance of occultism in the early modern Islamicate world, until recently assumed by scholars to have been a mark of terminal cultural decline—the intense occultism of the European Renaissance being an exception to this rule, of course. See *e.g.* Armand Abel, "La place des sciences occultes dans la décadence," in *Classicisme et déclin culturel dans l'histoire de l'Islam*, ed. Robert Brunschvig and Gustav Edmund von Grunebaum, Paris, Éditions Besson, 1957, p. 291-311. An index of the popular vitality of the Persianate geomantic tradition is its presence in the *Thousand and One Nights*; for a list of the relevant tales see Savage-Smith and Smith, "Islamic Geomancy," p. 220, n. 40.

37 MS Majlis 12534, p. 3; see Melvin-Koushki, "Persianate Geomancy."

and astronomer-astrologer and founder of the Maragha Observatory, wrote one of his two works on geomancy in both Arabic and Persian versions at the request of his Mongol patron Hülegü (r. 654/1256-663/1265), and is frequently cited as an authority in later Persian geomantic works.[38] With al-Ṭūsī as precedent, geomancy went on to exercise some of the best minds of the early modern Persianate world as a mainstream occult-scientific tradition.

This Arabo-Persian geomantic tradition constitutes Yazdī's immediate frame of reference, and also serves to contextualize Ibn Ḫaldūn's comments on the science inasmuch as it represents Islamicate geomancy in its mature phase, in this as in other fields marked by the refinement, systematization and expansion of western, Arabic texts by eastern writers from the 7th/13th century onward. Ibn Ḫaldūn himself unhappily admits the prevalence of geomantic practice "in all civilized lands"[39]—an enviable status that, by all accounts, only improved after the 8th/14th century, particularly in the Persianate east, where the science's popularity crescendoed in the runup to the Islamic millennium (1592 CE). How, then, to explain its ubiquity among the intellectual and political elite over the course of many centuries?

It must first be noted that geomancy was classified as an applied mathematical science from at least the 6th/12th century—a simple but oft-forgotten fact that explains its enduring appeal to philosophers, astronomers and mathematicians, including in the first-place prominent thinkers like Naṣīr al-Dīn al-Ṭūsī and Šams al-Dīn Ḫafrī (d. 942/1535).[40] This appeal, in turn, signals its

38 Savage-Smith, "Geomancy in the Islamic World"; Savage-Smith and Smith, "Islamic Geomancy," p. 216. The Persian version is typically titled *al-Wāfī fī ʿilm al-raml* and the Arabic *al-Sulṭāniyya fī l-raml*. Both versions are well represented in the manuscript record, particularly the Persian (preserved in at least 20 copies in Iran alone), and there is little reason to doubt the authenticity of the treatise's ascription to al-Ṭūsī. See Melvin-Koushki, "Persianate Geomancy."

39 Ibn Ḫaldūn, *Muqaddima*, I, p. 178.

40 Unlike "philosopher" (*faylasūf*), "astronomer" and "mathematician" are here used as terms of convenience: the former denotes a specialist in *ʿilm al-hayʾa*, while the latter only emerged as a professional term relatively late, and was infrequently used (my thanks to Sonja Brentjes for this observation). A prominent astronomer-philosopher under the early Safavids, Ḫafrī produced two treatises on geomancy, an otherwise untitled *Risāla dar Raml* (of which at least 20 manuscript copies survive; see *e.g.* MS Majlis 3931/3, p. 128-166, and Princeton MS New Series 1177/2, ff. 18b-27a) that emphasizes the philosophical-scientific framework of the science, and *Dawāzdah maṭlaʿ* (see *e.g.* MS Majlis 12509/1 ff. 1b-5a). On Ḫafrī's contributions to philosophy see Firouzeh Saatchian, *Gottes Wesen, Gottes Wirken: Ontologie und Kosmologie im Denken von Šams-al-Dīn Muḥammad al-Ḫafrī* (*gest. 942/1535*), Berlin, Klaus Schwarz ("Islamkundliche Untersuchungen", 305), 2011; on his signal importance to the history of astronomy see George Saliba, "A Sixteenth-Century Arabic Critique of Ptolemaic Astronomy: The Work of Shams al-Dīn al-Khafrī," *Journal for*

importance to the history of science more generally. (Lynn Thorndike treats of the science frequently in passing in his monumental *A History of Magic and Experimental Science*[41].) Most significantly, the mathematicalization of geomancy, together with lettrism, was part of a larger trend in Persianate intellectual history that culminated with the fateful mathematization of astronomy in the 9th/15th century by the members of the Samarkand Observatory, and drove the renaissance of neopythagoreanism in Iran through the Safavid period.[42]

This process began with Faḫr al-Dīn al-Rāzī (d. 606/1210), the great Ašʿarī philosopher-theologian and committed occultist; his *Ǧāmiʿ al-ʿulūm* (aka *Sittīnī*), the most comprehensive and innovative of the early Persian encyclopedias, is the first to formally elevate geomancy from the natural sciences to the mathematical. Specifically, it follows the standard Avicennan epistemological progression in treating the natural sciences (*ṭabīʿiyyāt*), the mathematical sciences (*riyāḍiyyāt*) and metaphysics (*ilāhiyyāt*) in that order, and under the rubric of the mathematical sciences includes geomancy (*raml*), together with geometry (*handasa*), mensuration (*masāḥa*), mechanics (*ǧarr al-aṯqāl*), war machines (*ālāt al-ḥurūb*), Indian arithmetic (*ḥisāb al-Hind*), mental calculation (*al-ḥisāb al-hawāʾī*), algebra (*al-ǧabr wa-l-muqābala*), arithmetic (*ariṯmāṭīqī*), magic squares (*aʿdād al-wafq*), optics (*manāẓir*), music (*mūsīqī*), astronomy (*hayʾat*), judicial astrology (*aḥkām* [*al-nuǧūm*]) and astral magic (*ʿazāʾim*). While he devotes some five pages to our divinatory art, however, al-Rāzī's presentation of geomancy is so rudimentary as to be useless to a would-be practitioner, suggesting that it had not yet entered the intellectual mainstream in the Persianate world.[43]

The *Nafāʾis al-funūn fī ʿarāʾis al-ʿuyūn* of Šams al-Dīn Āmulī (d. 753/1352), sometime *mudarris* at Sultaniyya under Öljeytü (r. 693/1294-706/1307),

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the History of Astronomy, 25 (1995), p. 15-38. Significantly for our purposes here, Saliba identifies Ḫafrī as the most important planetary theorist of the 10th/16th century, and believes him to be the first thinker to redefine mathematics "as a descriptive language of scientific processes," foreshadowing "the modern conception of the role of mathematics in science in general" ("Ḵafri, Šams-al-Din," *Encyclopaedia Iranica*). While Saliba's celebration of Ḫafrī is expressly and problematically teleological and positivist, I rely on it here to emphasize the Shirazi thinker's *neopythagorean* bent—wholly elided in Saliba's account; see Matthew Melvin-Koushki, "Powers of One: The Mathematicalization of the Occult Sciences in the High Persianate Tradition," *Intellectual History of the Islamicate World*, 5/1 (2017), p. 127-199.

41 Lynn Thorndike, *A History of Magic and Experimental Science*, New York, Columbia University Press, 1923-1958; see *e.g.* II, chap. 39.

42 So I argue in Melvin-Koushki, "Powers of One."

43 *Ǧāmiʿ al-ʿulūm* (*'Sittīnī'*), ed. Sayyid ʿAlī Āl-i Dāwūd, Tehran, Ṯurayyā, 1382/2003, p. 432-437. In this list geomancy, the 51st science, follows judicial astrology and precedes astral magic.

represents a second watershed in the Islamicate encyclopedic tradition.[44] It is the most extensive and polished *taṣnīf al-ʿulūm* work produced in either Arabic or Persian up to the mid-8th/14th century, and as such came to serve as the primary model for subsequent Persian encyclopedias.[45] The *Nafāʾis al-funūn* has as its own model al-Rāzī's *Ǧāmiʿ al-ʿulūm*, but it is far more complete and organized than that pioneering work. Most significantly, Āmulī, unlike al-Rāzī, questions the standard Avicennan hierarchy of sciences so forcefully reiterated a century prior by Quṭb al-Dīn al-Šīrāzī (d. 710/1311) in his *Durrat al-tāǧ li-ġurrat al-dabbāǧ*, at times even challenging the latter directly by counterposing what he asserts to be a more consistent classification system and moving toward a new conception of knowledge.[46] At the same time, he concurs with the Maragha philosopher-astronomer in classifying all occult sciences, including alchemy (*kīmiyā*), letter magic (*sīmiyā*), oneiromancy (*taʿbīr*), physiognomy (*firāsat*) and astrology, as derivative natural sciences—with one pointed and telling exception: geomancy. While al-Rāzī was the first encyclopedist to treat this form of "terrestrial astrology" as an applied mathematical science (*i.e.* an instance of the *furūʿ-i riyāḍī*) associated with astronomy, then, Āmulī goes further: he dissociates the science from astrology on the one hand and all other divinatory sciences on the other.[47] Though his motive for making it is not clear, Āmulī's move here may be considered an index of geomancy's increase in prestige in Iran during the Ilkhanid period. Moreover, at 20 pages Āmulī's discussion of geomancy is considerably fuller than his model's and detailed enough to serve as a basis for elementary practice.

The prestige of geomancy only continued to increase in the Persianate world from this point forward, such that by the 10th/16th century geomancy was considered an essential and uncontroversial application of the *quadrivium* in its Avicennan formulation. Writing at the turn of the Islamic millennium (1001/1593), or some two and a half centuries later, for the benefit of the

44 Šams al-Dīn Āmulī, *Nafāʾis al-funūn fī ʿarāʾis al-ʿuyūn*, ed. Abū l-Ḥasan Šaʿrānī and Ibrāhīm Miyānǧī, Tehran, Intišārāt-i Islāmiyya, 1389/2010.

45 Živa Vesel, *Les encyclopédies persanes : essai de typologie et de classification des sciences*, Paris, Recherche sur les civilisations ("Bibliothèque iranienne", 31; "Recherche sur les civilisations. Mémoire", 57), 1986, p. 39-42.

46 *Ibid.*, p. 41; Melvin-Koushki, "Powers of One."

47 Šams al-Dīn Āmulī, *Nafāʾis al-funūn*, III, p. 537-556. Here again I use the designation "applied mathematical science" as a term of convenience; it is not to be understood in its strict 19th-century European sense.

Mughal Emperor Akbar, in his *Qawāʿid al-hidāya*, Hidāyat Allāh Munağğim-i Šīrāzī programmatically states:[48]

> Know (God guide you) that philosophy (*ḥikmat*) is divided into three categories: the natural sciences (*ʿilm-i ṭabīʿī*), the mathematical sciences (*ʿilm-i riyāḍī*) and metaphysics (*ʿilm-i ilāhī*). The principal mathematical sciences are four: astronomy (*hayʾat*), arithmetic (*ʿadad*), geometry (*handasa*) and music (*mūsīqī*). The mathematical disciplines derived from these are numerous, and include geomancy (*raml*) and astrology (*nuǧūm*), which are exceedingly noble and useful sciences.[49]

At the same time, geomancy's focus on the four elements—fire, air, water, earth—also gave it a strong claim to being a natural science. As the *Šağara u ṯamara*, a popular Persian geomantic manual of the period (formally a commentary on a Persian translation of al-Zanātī's *Šağara*), warns:

> The modality of this science is a difficult one, for it is a physiological science (*ʿilm-i ṭabāʾiʿ*); whoever does not have a firm grasp of physiology will not be able to use this science.[50]

Our divinatory science, in other words, occupied the epistemological intersection between astrology and alchemy.[51] It is significant in this context that the conception of the four-cornered *nuqṭa* as primordial embodiment of the four elements was likewise central to Maḥmūd Pasīḫānī's (d. 831/1428) reformulation of the Ḥurūfism of his erstwhile teacher Faḍl Allāh Astarābādī (d. 796/1394), and drove what was seen by critics to be the thoroughgoing naturalism or materialism of Nuqṭawī doctrine.[52]

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48 The *Qawāʿid al-hidāya* was first written for Akbar in 962/1555 just before his accession and reworked in 1001/1593. Storey, *Persian Literature*, II/3, p. 482, nº 850 (4, 5). On its status as slightly edited version of a Safavid geomantic manual see Melvin-Koushki, "Persianate Geomancy."

49 Hidāyat Allāh Munağğim-i Šīrāzī, *Qawāʿid al-hidāya*, MS Majlis 12563, p. 312.

50 *Ibid.*, p. 820.

51 Klein-Franke, "The Geomancy of Aḥmad b. ʿAlī Zunbul," p. 33, 35.

52 See *e.g.* Ṣādiq Kiyā's classic study of the movement, *Nuqṭawiyān yā Pasīḫāniyān*, Tehran, Anğuman-i Īrān, 1320/1941. In his *Durr al-yatīm*, for instance, Muḥammad Dihdār (d. 1016/1607), son of the important Safavid lettrist author Maḥmūd Dihdār Šīrāzī (*fl.* 984/1576), classes the Nuqṭawīs as an execrable subset of those he calls the mass of deficient naturalist-materialists (*ǧumhūr-i ṭabīʿiyyīn u nāqiṣān-i dahriyān*). Muḥammad Dihdār, *Rasāʾil-i Dihdār*, ed. Muḥammad Ḥusayn Akbarī Sāwī, Tehran, Nuqṭa, 1375/1996,

Geomancy's status as a *mathematical* occult science appears to have become particularly salient by the early 9th/15th century, when it began to be explicitly associated with lettrism (*ʿilm al-ḥurūf*), particularly letter magic (*sīmiyāʾ*) and letter divination (*ǧafr*), both based on number theory (*ʿilm al-ʿadad*). This trend is exemplified in Luṭf Allāh b. ʿAbd al-Malik Nīšāpūrī Samarqandī's[53] *Ḫulāṣat al-baḥrayn*, written in 812/1409, which treats of geomancy and magic squares (*aʿdād-i wafq*) (the two seas of the title) as complementary sciences.[54] His explicit thesis: geomancy tells you the future, and if you don't like what you hear, you can always change the future with letter magic. This prospect, very simply, explains the attractiveness of this and other occult sciences to ruling elites throughout the early modern Islamicate world and their heavy patronage of the same; used in conjunction with lettrism in particular, geomancy promises the direct control of one's political and material fate.[55]

I have shown elsewhere that lettrism was reformulated during this same early 9th/15th-century moment as the primary vehicle of Islamicate neopythagoreanism and philosophical occultism, and advanced as a superior

p. 131. He criticizes their doctrine as follows (*ibid.*, p. 141): "These [people] who associate themselves with the dot (*nuqṭa*) mean by it the elemental particles (*aǧzā-yi ʿunṣurī*) that are in the dot; and their sublimest (*ḥaqīqī*) dot is the center of the earth (*markaz-i ḫāk*). For this reason, whenever their leader [*i.e.* Maḥmūd Pasīḫānī] would mention his own name in his treatises he would say 'I who have dirt on my head'—that is to say, 'I who am the center of the earth'! [The Nuqṭawīs aside], however, the science of the dot (*ʿilm-i nuqṭa*) is a noble science; for the dot refers to the absolute aspect (*ḥayṯiyyat-i muṭlaqa*). The treatise *Secrets of the Dot* (*Asrār al-nuqṭa*) by Mīr Sayyid ʿAlī Hamadānī (may He sanctify his spirit) is well-known [in this connection]. And according to the the felicitating speech of the holy Commander of the Faithful [ʿAlī b. Abī Ṭālib] on [the subject of] the dot, 'Knowledge is a dot multiplied by the ignorant,' here referring to the ultimate reality (*ḥaqīqat al-ḥaqāyiq*) or first entification (*taʿayyun-i awwal*)."

53 He is perhaps to be identified with the celebrated poet Luṭf Allāh b. Sulaymān Šāh Nīšāpūrī (d. 812/1409), a Khurasan-based encomiast to the Sarbidars and Timurids known for his ascetic character and expertise in astrology. See *Dīwān-i Luṭf Allāh Nīšābūrī*, ed. Rasūl Ǧaʿfariyān, Tehran, Asnād-i maǧlis-i šūrā-yi islāmī, 1390/2011, p. 117.

54 See Storey, *Persian Literature*, II/3, p. 481, nº 847. Tellingly, a copy of his teacher ʿAbd al-Ġanī Ḥāfiẓ Šīrwānī's *Anwār al-raml* is bound together with an anonymous treatise on magic squares, *Kayfiyyat-i aʿmāl-i aʿdād-i wafq*, in MS Marʿašī 7142.

55 As Maḥmūd Kāšī, *munaǧǧim* to the Timurid Iskandar Sulṭān (on whom see below), argues for the necessity of astrology: "The wise soul manages the actions of the heavens, just as the farmer manages the power of nature by ploughing and watering." Laurence P. Elwell-Sutton, "A Royal Timurid Nativity Book," in *Logos Islamikos: Studia Islamica in Honorem Georgii Michaelis Wickens*, ed. Roger M. Savory and Dionisius A. Agius, Toronto, Pontifical Institute of Mediaeval Studies, 1984, p. 130.

alternative to both mainstream peripatetic-illuminationist philosophy and sufi theory.[56] Significantly, the network of thinkers who accomplished this reformulation appear to have called themselves the Iḫwān al-Ṣafā' in conscious reference to the shadowy group of 4th/10th-century neopythagorean-occultist encyclopedists of the same name. By the early 9th/15th century this network included most prominently Šayḫ Badr al-Dīn of Simavna (d. between 819/1416 and 823/1420) and ʿAbd al-Raḥmān al-Bisṭāmī (d. 858/1454) in Anatolia,[57] and Ṣā'in al-Dīn Turka Iṣfahānī (d. 835/1432) and our historian Šaraf al-Dīn Yazdī in western Iran, with Sayyid Ḥusayn Aḫlāṭī (d. 799/1397), famed Tabrizi Kurdish occultist and personal physician-alchemist to Sulṭān Barqūq, standing as the network's pivot.[58] Ibn Turka, in turn, stands as its leading theoretician in the east, being responsible for the systematization of this philosophical lettrism[59]—the basis of his student and friend Yazdī's historical and occult-scientific outlook. That this intellectual confraternity's interests were not confined to lettrism is suggested by the fact that another of Aḫlāṭī's disciples, Ḥasan Abarqūhī, was known strictly as a geomancer,[60] and Aḫlāṭī himself has been credited as author of the seminal Persian geomantic manual *Risāla-yi Surḫāb*.[61]

As a divinatory methodology based on number theory, then, geomancy fit neatly into the newly resurgent neopythagorean-neoplatonic (*fīṯāġūrī-aflāṭūnī*) project that was sweeping the early 9th/15th-century intellectual scene, and served as the primary foundation for the philosophical-scientific

56 See Melvin-Koushki, "The Quest."

57 Cornell H. Fleischer, "Ancient Wisdom and New Sciences: Prophecies at the Ottoman Court in the Fifteenth and Early Sixteenth Centuries," in *Falnama: The Book of Omens*, ed. Massumeh Farhad and Serpil Bağcı, London-Washington, Thames & Hudson-Freer Gallery of Art, 2009, p. 232-243: p. 235; İhsan Fazlıoğlu, "İlk Dönem Osmanlı İlim ve Kültür Hayatında İhvanu's-Safâ ve Abdurrahman Bistâmî," in *Dîvân: İlmî Araştırmalar*, 2 (1996), p. 229-240; Denis Gril, "Ésotérisme contre hérésie: ʿAbd al-Rahmân al-Bistāmī, un représentant de la science des lettres à Bursa dans la première moitié du XVe siècle," in *Syncrétismes et hérésies dans l'Orient seldjoukide et ottoman* (*XIVe-XVIIIe siècle*), ed. Gilles Veinstein, Louvain-Paris, Peeters ("Collection Turcica", 9), 2005, p. 183-195; see Melvin-Koushki, "The Quest," p. 16-19.

58 On Aḫlāṭī see Binbaş, *Intellectual Networks*, p. 114-140; Melvin-Koushki, "The Quest," p. 45-57.

59 His *Kitāb al-Mafāḥiṣ* or *Book of Inquiries* in particular is a *summa* of philosophical lettrism and Islamic neopythagoreanism; see Melvin-Koushki, "The Quest," p. 330-378.

60 Ṭabāṭabā'ī Yazdī identifies al-Ḥāǧǧ Ḥasan as a student of Sayyid Ḥusayn Aḫlāṭī, and cites him as a geomantic authority without associating him with a specific work (*Nafaḥāt al-asrār*, p. 29).

61 Sayyid Muḥsin al-Amīn, *Aʿyān al-šīʿa*, ed. Ḥasan al-Amīn, Beirut, Dār al-taʿāruf, 1986-2006, VI, p. 94.

and spiritual-political adventurousness that characterized the early modern Persianate world; this conceptual confluence is plainly reflected in Luṭf Allāh Nīšāpūrī's innovative marriage of geomancy and letter magic on the one hand and Yazdī's pointedly lettrist defense of the science on the other.

So far geomancy's scientific classification; but what of its primary mechanism? Geomancy, like all occult sciences as practiced in the early modern Islamo-Christianate world, is predicated on the hermetic principle of correspondence (*munāsaba*): as above, so below. This union of supernal and infernal, of spirit and matter, of eternity and time is governed by the law of (gendered) attraction or love (*ḥubb*, *maḥabba*, *ʿišq*) obtaining between all bodies, from planetary to atomic.[62] The neoplatonists hold that these two principles erect a hierarchy of being that may be ascended and descended at will; the neopythagoreans hold that number is the only shuttle capable of returning

62 *I.e.* the Love and Strife of Empedocles, fused with the erotic motion of Plotinus. This doctrine was fully islamicized by the 4th/10th century, when the Brethren of Purity declared in their 37th treatise on love, embracing the Plotinian position: "Divine wisdom and solicitude has bound all existents together with a single bond and strung them together in a single arrangement"; "God is the First Beloved, and the celestial sphere revolves solely out of longing (*šawq*) for Him and love (*maḥabba*) for permanence." Iḫwān al-Ṣafāʾ, *Rasāʾil Ikhwān al-ṣafāʾ*, Beirut, al-Aʿlamī, 1426/2005, III, p. 227, 234. Likewise, in his *Treatise on Love* (*Risāla fī l-ʿišq*), even Ibn Sīnā (d. 428/1037), second father of peripateticism, posits love as a primary ground of being, and intrinsic to the divine essence: "Love is the manifestation of both essence (*al-ḏāt*) and existence (*al-wuǧūd*), that is, in the Good (*al-ḫayr*); the being of existents is therefore either by reason of the love in them [for the Good], or identical to love itself." Ibn Sīnā, *Ǧāmiʿ al-badāʾiʿ*, ed. Muḥammad Ḥasan Ismāʿīl, Beirut, Dār al-kutub al-ʿilmiyya, 1425/2004, p. 67. In the post-Mongol period, however, the philosophical doctrine of love as ground of being came to be most closely associated with Ibn ʿArabī (d. 638/1240), who embraced the sufi concept of *al-nikāḥ al-sārī fī ǧamīʿ al-ḏarārī* ("the marriage act pervading all atoms"); the same concept is referred to by his lettrist expositor Ibn Turka as "perpetual bonding" (*al-ʿaqd al-sāʾir*) and "pervasive coupling" (*al-nikāḥ al-sārī*). Melvin-Koushki, "The Quest," p. 364.

Most famously the basis for Isaac Newton's (d. 1727) theory of gravitation, this doctrine was ubiquitous in early modern thought, from the political philosophy of Ǧalāl al-Dīn Dawānī (d. 908/1502) in the 9th/15th century to the millenarian empiricism of Francis Bacon (d. 1626) in the early 11th/17th. It is thus highly significant in this connection that Bacon too elevated *artificial divination*—considered far inferior to natural divination from Plato onward, and epitomized in the Islamo-Christianate world by such techniques as geomancy—to the status of an experimental natural science, pointedly renaming it natural divination to this end. See A.P. Langman, "The Future Now: Chance, Time and Natural Divination in the Thought of Francis Bacon," in *The Uses of the Future in Early Modern Europe*, ed. Andrea Brady and Emily Butterworth, New York, Routledge ("Routledge studies in Renaissance literature and culture", 12), 2010, p. 142-158, esp. p. 152.

the human intellect to its transcendent, unitary origin. Properly manipulated, then, the primordial virtue of number may be activated in a way that grants knowledge of the unseen—past, present and future, essences and secret thoughts. Geomancy is one such method of number manipulation whose efficacy depends on the correspondence of above to below along the neoplatonic Chain of Being.[63]

Most importantly for the history of science, this neopythagorean-neoplatonic framework underlying all occultist practice is itself predicated on the principle of secondary causality vertically cascading down that Chain, though in a form more reminiscent of the spooky—*i.e.* occult—causality that makes possible quantum-mechanical nonlocality. Indeed, perhaps *munāsaba* itself is better translated as 'entanglement'—or "synchronicity," as C.G. Jung, who was much preoccupied by the study of the *I Ching* and the occult sciences generally, famously termed the primary mechanism of divination, positing number as the archetype that marries mind to matter.[64] As that may be, geomancy's basic thesis that a binary code underlies and shapes physical reality has long been a staple idea among logocentric metaphysicians, Muslim, Jewish and Christian alike, a host of whom embraced the Danielic art.

It is on this point that Yazdī takes to task his anti-occultist interlocutor, Ibn Ḫaldūn, who heaps scorn and censure on those who investigate secondary causes,[65] by counterposing geomancy as a science precisely devoted to such investigation. Yet the Tunisian historian cannot simply be dismissed as an occasionalist; it is painfully ironic, to say the least, that he inveighs against the investigation of secondary causes in a masterwork devoted precisely to investigating the causes behind the rise and fall of civilizations.[66]

In geomancy, then, we have a mainstream, mathematical-natural science predicated on a neopythagorean-neoplatonic system and animated by the twin principles of correspondence and secondary causation that was of enduring interest to scholarly and ruling elites throughout the Islamo-Christianate world during the 7th/13th-10th/16th-century moment most especially, and is

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63 This point is strongly made by Ḫafrī, for instance, in his *Risāla dar Raml* (MS Majlis 3931/3, p. 128-129; Princeton MS New Series 1177/2, ff. 18b-19a). On the Chain of Being theory generally see Arthur O. Lovejoy, *The Great Chain of Being: A Study of the History of an Idea*, Cambridge, Harvard University Press, 1936; on the same in Islamicate thought see Aziz Al-Azmeh, *Arabic Thought and Islamic Societies*, London, Croom Helm ("Exeter Arabic and Islamic series", 1), 1986, p. 2-13; on the same in the *Muqaddima* see *id., Ibn Khaldūn*, p. 66-67; Asatrian, "Ibn Khaldūn," p. 77-79.

64 See *e.g.* the study by his student Marie-Luise von Franz, *On Divination and Synchronicity*.

65 Ibn Ḫaldūn, *Muqaddima*, III, p. 23-27; *id., The Muqaddimah*, III, p. 34-37.

66 My thanks to Justin Stearns for this observation.

still in common use to the present. But Ibn Ḫaldūn begged to differ—albeit in rather uncertain terms. We therefore turn to his harsh but subtle critique of the occult sciences in general and geomancy in particular.

Ibn Ḫaldūn Against the Occult Sciences

In openly attacking the occult sciences the Tunisian historian was hardly the odd scholar out. Anti-occultist polemics, featuring in the Arabic textual tradition from its very beginning in the 2nd/8th century, had built up a venerable pedigree by the late 8th/14th century; it is one of the few subjects where philosophers like Ibn Sīnā (d. 428/1037), theologians like al-Ġazālī (d. 505/1111), traditionalists like Ibn Taymiyya (d. 728/1328) and historians like Ibn Ḫaldūn find common cause.[67] Nor was the latter original in singling out astrology and alchemy as the worst offenders against religion and reason; Ibn Sīnā and Ibn Taymiyya's student Ibn Qayyim al-Ǧawziyya (d. 751/1350), Ibn Ḫaldūn's earlier contemporary, had both done the same. Indeed, though a direct connection is difficult to show, Ibn Ḫaldūn's immediate precedent was likely Ibn al-Qayyim's sustained anti-occultist polemic in his *Miftāḥ dār al-saʿāda* (*Key to the Abode of Felicity*), which systematically attacks the occult sciences, particularly astrology, alchemy, magic and various forms of divination, using

67 Yahya Michot helpfully compiles a list of anti-astrology polemicists as follows in his "Ibn Taymiyya on Astrology: Annotated Translation of Three Fatwas," *Journal of Islamic Studies*, 11/2 (2000), p. 147-208: p. 151: al-Ḫalīl b. Aḥmad (d. *ca* 170/786), al-Šāfiʿī (d. 204/820), Abū Tammām (d. *ca* 231/845), Ṯābit b. Qurra (d. 288/901), Ḥasan b. Mūsā al-Nawbaḫtī (d. *ca* 310/920), Abū l-Ḥasan al-Ašʿarī (d. 324/935), al-Fārābī (d. 339/950?), Abū l-Qāsim ʿĪsā b. ʿAlī (d. 391/1001), Uqlīdisī (mid-4th/10th c.), al-Bāqillānī (d. 403/1013), Abū Ḥayyān al-Tawḥīdī (d. 414/1023?), Ibn Sīnā (d. 428/1037), Ibn al-Hayṯam (d. 430/1039), al-Bīrūnī (d. after 442/1050), Ibn Ḥazm (d. 456/1064), al-Ḫaṭīb al-Baġdādī (d. 463/1071), al-Ġazālī (d. 505/1111), Ibn Rušd (d. 595/1198), al-Samawʾal al-Maġribī (*fl.* 6th/12th c.), Abū l-Barakāt al-Baġdādī (d. after 560/1164), Ibn Taymiyya (d. 728/1328), Ibn Qayyim al-Ǧawziyya (d. 751/1350), Ḫalīl al-Ṣafadī (d. 764/1363), Ibn al-Šāṭir al-Dimašqī (d. 777/1375), Muḥammad al-Damīrī (d. 808/1405), Ibn Ḫaldūn (d. 808/1406) and Ibn Ḥağar al-Haytamī (d. 974/1567). Ibn Taġrībirdī (d. 874/1470), who delighted in fleering astrologers, should be added to this list (Irwin, "Al-Maqrīzī and Ibn Khaldūn," p. 225-226), as should Qāḍī ʿAbd al-Ǧabbār (d. 415/1025) and Moses Maimonides (d. 601/1204). See George Saliba, "Islamic Astronomy in Context: Attacks on Astrology and the Rise of the *Hayʾa* Tradition," *Bulletin of the Royal Institute for Inter-Faith Studies*, 4/1 (2002), p. 37-41; Y. Tzvi Langermann, "Maimonides' Repudiation of Astrology," *Maimonidean Studies*, 2 (1991), p. 123-158. For Ibn Rušd's critique of occultism more generally see his *Tahāfut al-Tahāfut*, ed. M. Bouyges, Beirut, Imprimerie catholique, 1930, p. 509-511.

natural-scientific, theological and traditionalist arguments[68]—an obvious template for Ibn Ḫaldūn's own critique of occultism on theological, natural-scientific, social and political grounds.[69]

Yet Ibn al-Qayyim's polemic, for all its urbane learnedness, is clearly reactionary; that a strict Ḥanbalī jurist-theologian of his training found it necessary to recruit the help of natural philosophy and the mathematical sciences to this end smacks of desperation—Ibn Taymiyya would hardly have approved—,[70] and suggests that he saw the burgeoning popularity of occultism in his own day as a nigh-unstoppable force.[71] This impression is confirmed by Ibn al-Qayyim's facile identification of occultism with Ismaʿilism, paganism and philosophy in his *Iġāṯat al-lahfān min maṣā'id al-šayṭān* (*Saving the Sighing from the Snares of Satan*), a rhetorical move calculated to fan the flames of sectarian fear and theological outrage in his readership. In this work he traces a history of philosophy from the Greeks to the 8th/14th century, and asserts that while Socrates and his student and successor Plato were essentially proto-Muslim, Plato's own student Aristotle was responsible for the terminal perversion of philosophy; Aristotle's greatest exponent in the Islamic dispensation is of course Ibn Sīnā, a self-confessed Ismaʿili heretic (*kāna min al-qarāmiṭa l-bāṭiniyya*), whose school now dominates the philosophical scene—monsters of *taʿṭīl* all. During the same period, the infamously occultist *Rasā'il* of the Iḫwān al-Ṣafā' were written for precisely this audience of heretics (*zanādiqa*, *malāḥida*). Ibn

68 See John Livingston, "Science and the Occult in the Thinking of Ibn Qayyim al-Jawziyya," *Journal of the American Oriental Society*, 112/4 (1992), p. 598-610.

69 It should be noted that previous arguments against the occult sciences were more often made on purely religious grounds rather than natural-scientific; see *e.g.* al-Nawawī, *Fatāwā l-Imām al-Nawawī*, ed. Maḥmūd al-Arnā'ūṭ, Damascus, Dār al-fikr, 1999, p. 132-135; Ḍiyā' al-Dīn b. al-Aṯīr, *al-Maṯal al-sā'ir fī adab al-kātib wa-l-šāʿir*, ed. Muḥammad Muḥyī l-Dīn ʿAbd al-Ḥamīd, Cairo, Maṭbaʿat Muṣṭafā l-Bābī l-Ḥalabī, 1938-1939, II, p. 148-149.

70 Indeed, Ibn al-Qayyim even goes so far as to invoke the philosophers' critiques of astrology—this while fulminating against philosophy and logic on theological and sectarian grounds in the *Miftāḥ* and elsewhere. See Ibn al-Qayyim, *Miftāḥ dār al-saʿāda*, ed. ʿAbd al-Raḥmān b. Ḥasan b. Qā'id, Mecca, Dār ʿālam al-fawā'id, 1432/2011, p. 1463.

71 As Livingston observes: "Ibn al-Qayyim's reliance on arguments drawn from the exact sciences and natural philosophy, curious defenses considering the theological conceptions of the Ḥanbalī traditionist, is a measure of how threatening he perceived the popularity of the occult sciences to be." Livingston, "Science and the Occult," p. 599. On Ibn al-Qayyim's complicated relationship to the rational sciences see Y. Tzvi Langermann, "The Naturalization of Science in Ibn Qayyim al-Ǧawziyyah's *Kitāb al-Rūḥ*," *Oriente Moderno*, 90/1 (2010), p. 203-220.

al-Qayyim here blasts Naṣīr al-Dīn al-Ṭūsī—or as he calls him, Naṣīr al-Širk wa-l-Kufr—in particular, whose many crimes against Islam include assisting Hülegü in the destruction of the caliph, judges, jurists and traditionists while striving to ensure the survival of philosophers (*falāsifa*), astrologers (*munağğimūn*), naturalists (*ṭabāʾiʿiyyūn*) and mages (*saḥara*), and plundering the endowments of mosques, madrasas and *ribāṭ*s for the benefit of his depraved associates; he also took up the study of magic toward the end of his life, becoming an idol-worshiping sorcerer in his own right. The moral of Ibn al-Qayyim's declinist narrative: whenever and wherever Jews, Christians or Muslims have pursued the study of philosophy and logic, those gateway drugs to occultism, thereby departing from the revealed truths of scripture, God has punished them with conquest and captivity—witness the Ismaʿili ascendancy in the west, the Christian takeover of al-Andalus and the Tatar invasion of the east. In other words, Ibn al-Qayyim holds the popularity of Avicennan peripatetic philosophy to be directly responsible for the Mongol conquest of western Asia, which in turn allowed for the burgeoning of occultism through the intermediation of traitors like al-Ṭūsī.[72]

Ibn al-Qayyim's arguments in both the *Miftāḥ* and the *Iġāṯa* thus imply the following two-stage scenario: Ismaʿilism and philosophy represented the greatest threats to Sunni orthodoxy in the pre-Mongol period, which accordingly saw the emergence of great anti-Ismaʿili and anti-philosophy champions of Sunnism like al-Ġazālī (d. 505/1111). The Mongols' destruction of organized Ismaʿilism did not extinguish that pernicious heresy, however; it persists, viruslike and barely disguised, in contemporary occultism, which like Ismaʿili doctrine is based on neopythagorean-neoplatonic (*fīṯāġūrī-aflāṭūnī*) philosophy.[73] And their patronage of peripatetic philosophy and occult practice in the person of Naṣīr al-Dīn al-Ṭūsī only exacerbated the situation—the popularity of philosophy, after all, was what called down the Mongol scourge in the first place. As such, occultism and philosophy now represent the greatest threats to Sunni orthodoxy in the post-Mongol period, with Ibn al-Qayyim, al-Ġazālī-like, standing in the breach.[74]

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72 Naṣīr al-Dīn al-Ṭūsī, *Iġāṯat al-lahfān*, ed. Muḥammad Sayyid Kīlānī, Cairo, Maṭbaʿat Muṣṭafā l-Bābī l-Ḥalabī, n.d., p. 260-266.

73 As Livingstone observes, for Ibn al-Qayyim, occultism, as a form of "Ismāʿīlī Pythagorean gnosticism," was the "superstitious offspring of legitimate science and Ismāʿīlī spiritual alchemy." Livingston, "Science and the Occult," p. 608.

74 *Ibid.*, p. 598-600, 608. By this standard, of course, occult philosophers like Ibn Turka, who systematized lettrism as a superior metaphysics, must be considered a double threat to society and religion.

Writing some decades later, Ibn Ḫaldūn, a Mālikī jurist with Ašʿarī leanings, is no less reactionary in this respect; he too seeks to defend the ramparts of religion with dire warnings as to the danger occultism poses for human civilization in general and believers and political actors in particular. And like Ibn al-Qayyim, he stands appalled witness to the renaissance of occultism in the 8th/14th century. But the rising tide was not to be stemmed. Ibn Ḫaldūn here appears not as man ahead of his time but as a conservative, fideistic voice, a historian run aground on the wrong side of history, his warnings left unheeded by early modern dynasts eager to use the occult sciences as a tool for political advancement—and a basis for their increasingly frequent claims to *millennial sovereignty*.[75] Significantly, his appears to be the last important attack on occultism as such in the premodern Arabic scholarly tradition.[76] Even Ibn

75 On this theme see *e.g.* Azfar Moin, *The Millennial Sovereign: Sacred Kingship and Sainthood in Islam*, New York, Columbia University Press, 2012; Matthew Melvin-Koushki, "Early Modern Islamicate Empire: New Forms of Religiopolitical Legitimacy," in *The Wiley-Blackwell History of Islam*, ed. Armando Salvatore, Roberto Tottoli and Babak Rahimi, Malden, Wiley-Blackwell, forthcoming 2017. Alexander Knysh identifies the Akbarian seal of the saints doctrine—espoused by Ibn Turka and Yazdī, who rendered it (occult-) scientific—as particularly politically problematic for the courtier and statesman Ibn Ḫaldūn in its status "as a potential banner and ideology for violent messianic uprisings." Alexander Knysh, *Ibn ʿArabi in the Later Islamic Tradition: The Making of a Polemical Image in Medieval Islam*, Albany, State University of New York Press, 1999, p. 198. He concludes (*ibid.*, p. 195): "Plainly, sociopolitical rather than theological considerations motivated Ibn Khaldun's severe criticism of the messianic theories imputed to Ibn ʿArabi and to other monistic Sufis […]. Ibn Khaldun's misgivings should be seen against the background of the turbulent Maghribi history that was punctuated by popular uprisings led by self-appointed *mahdi*s who supported their claims through magic, thaumaturgy, and occult prognostication."

76 This is not to suggest that occultism did not continue to earn the ire of some Muslim scholars through the early modern period, only that polemic seems to have been focused elsewhere (in 11th-12th/17th-18th-century Safavid Iran, for instance, the preferred targets were organized sufism and Sunnism); I know of no sustained anti-occultism critique on the model of Ibn al-Qayyim's and Ibn Ḫaldūn's produced after the latter's death. In Iran, of course, the early modern occult waters were muddied by the anarchic energy of the Ḥurūfiyya and then the Nuqṭawiyya, this to the point that "Nuqṭawī" became a standard term of abuse during the early Safavid period—much like *naw-mulḥid* had been in the Timurid, particularly as used against occultists like Ibn Turka and Yazdī (my thanks to Daniel Sheffield for this observation). And compared to the enthusiatic embrace of occultists at the Ottoman and Mughal courts, Safavid patronage of occultism seems to have been somewhat lower-key; yet geomancers, astrologers and lettrists remained valued members of the Safavid court, and the intellectual elite in Fars pursued the occult sciences unchecked. See Matthew Melvin-Koushki, "The Occult Sciences in Safavid Iran and

Ḫaldūn's student and ardent admirer, the preeminent Mamluk historian Taqī l-Dīn al-Maqrīzī (d. 845/1442), begged to differ with his teacher on this point, and enthusiastically promoted the occult sciences,[77] particularly geomancy. The Mamluk historian al-ʿAynī (d. 855/1451), for instance, accuses him of being obsessed with *ḍarb al-raml*.[78] Indeed, in *al-Muqaffā*, al-Maqrīzī goes so far as to valorize Ḥasan-i Ṣabbāḥ (d. 518/1124) himself, attributing the dread Ismaʿili lord's political genius to his mastery of various occult sciences (*ġawāmiḍ ʿulūm*).[79]

For alongside the anti-occultist current that persisted to the end of the 8th/14th century there ran an equally persistent pro-occultist countercurrent driven by thinkers of a more neopythagorean-neoplatonic-monist bent.[80] This countercurrent is here represented by Ibn Ḫaldūn's respondent Yazdī, whose rebuttal may be considered a resounding success; if the enthusiastic (if sometimes fairweather) patronage of the occult sciences by Mamluk, Timurid, Aqquyunlu, Qaraquyunlu, Ottoman, Safavid, Mughal, Qutbshahi and other

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Safavid Occultists Abroad," in *The Safavid World*, ed. Rudi Matthee, New York, Routledge, forthcoming. As that may be, much further research will be necessary to determine the extent to which occultism was or was not bundled together with Ismaʿilism, Nuqṭawism and/or sufism by early modern Muslim polemicists.

77 Al-Maqrīzī was Ibn Ḫaldūn's student for at least 30 years, in *ʿilm al-mīqāt* specifically and historiography generally, and his *Ḫiṭaṭ* has been identified as ingeniously Ibn Ḫaldūnian. Nasser Rabbat, "Was al-Maqrīzī's *Khiṭaṭ* a Khaldūnian History?" *Der Islam*, 89/2 (2012), p. 118-140. It is ironic that Ibn Ḫaldūn's own interest in divination, especially the *zāʾirǧa* technique, seems to have piqued al-Maqrīzī's occultist interests. Irwin, "Al-Maqrīzī and Ibn Khaldūn," p. 230; Rabbat, "Was al-Maqrīzī's *Khiṭaṭ*," p. 120. Irwin remarks of al-Maqrīzī in the same article (p. 230): "History was at the core of his oeuvre, but occult and eschatological concerns were at the core of his history." Moreover, a number of stories exist that "reveal the interest both men shared in the power of the preternatural, especially when transmitted via prophecies, visions, and dreams, as a means of explaining the world no less important than observation and experience. This has prompted al-Saḫāwī, who disliked both men, to comment on their credulity by saying that al-Maqrīzī endeared himself to Ibn Ḫaldūn when he read his future and predicted the time of his judgeship, which happened as he predicted, and that was considered a marvel." Rabbat, "Was al-Maqrīzī's *Khiṭaṭ*," p. 121; see al-Saḫāwī, *al-Ḍawʾ al-lāmiʿ*, Cairo, Maktabat al-qudsī, 1935-1937, II, p. 24.

78 Al-ʿAynī, *ʿIqd al-ǧumān fī taʾrīḫ ahl al-zamān*, ed. ʿAbd al-Razzāq al-Ṭanṭāwī l-Qarmūṭ, Cairo, Maṭbaʿat ʿalāʾ, 1985, II, p. 574.

79 Al-Maqrīzī, *al-Muqaffā l-kabīr*, ed. Muḥammad al-Yaʿlāwī, Beirut, Dār al-ġarb al-islāmī, 1411/1991, III, p. 327-334, nº 1160; my thanks to Carol Hillenbrand for this reference.

80 Ibn al-Qayyim calls them *al-fīṯāġūriyya wa-l-aflāṭūniyya*, dismissing them as being but the plaything of Satan like a ball in the hands of a child. Ibn al-Qayyim, *Iġāṯat al-lahfān*, p. 264.

early modern ruling elites is any indication,[81] the latter's arguments were to prove the more convincing to scholars and sovereigns till at least the 11th/17th century, and in some cases the late 13th/19th.[82] That the *Muqaddima* exerted no detectable influence on Arabic historiography until the later Ottoman period is due in no small part, I argue, to its confused, off-putting and old-fashioned anti-occultist agenda, part and parcel of Ibn Ḫaldūn's general opposition to the philosophical sciences on the one hand and to the new forms of messianic kingship that would define post-Mongol Islamdom on the other.[83]

Ibn al-Qayyim's sectarian argument was neither new nor baseless, however; this neopythagorean-neoplatonic-monist countercurrent was indeed frequently associated withIsmaʿilism, particularly in the wake of the Iḫwān al-Ṣafāʾ (who despite being at most only semi-Ismaʿili in outlook were received most enthusiastically in Ismaʿili circles) and their embrace of the occult sciences within a neopythagorean-neoplatonic-aristotelian framework.[84] That early

81 On Aqquyunlu and Safavid patronage of lettrism, for example, see my forthcoming *The Occult Science of Empire in Aqquyunlu-Safavid Iran*; on Timurid patronage of the same see my forthcoming *Occult Philosophers and Philosopher Kings in Early Modern Iran*, a study of Ibn Turka and the New Brethren of Purity; on courtly patronage of occultism in the Ottoman Empire see the abovementioned studies by Cornell Fleischer and Ahmet Tunç Şen's article in this special issue; on the same in the Mughal Empire see Azfar Moin's *The Millennial Sovereign* and Daniel Sheffield's "The Lord of the Planetary Court: Cosmic Aspects of Millennial Sovereignty in the Thought of Āẕar Kayvān and His Associates" (forthcoming); and on the same in the Maghrib see H.P.J. Renaud, "Divination et histoire nord-africaine au temps d'Ibn Khaldūn," *Hespéris*, 30 (1943), p. 213-221.

82 On the survival of a specifically Timurid brand of imperial lettrism in 13th/19th-century Manghit Bukhara see Melvin-Koushki and Pickett, "Mobilizing Magic."

83 Ibn Ḫaldūn's anti-occultism was hardly the only issue, of course; he and his history were much abused by contemporary Mamluk historians for a variety of reasons. Most damningly, Ibn Ḥağar al-ʿAsqalānī (d. 852/1449) and al-Saḫāwī (d. 902/1497) derided his poor mastery of historical data, especially for regions beyond North Africa; his disjointed, superficial and hence unconvincing style; and his contrarian pigheadedness. It is moreover revealing that even al-Maqrīzī, the *Muqaddima*'s sole champion, declined to follow its method. Simon, *Ibn Khaldūn*, p. 18-19. After centuries of obscurity, however, the *Muqaddima* enjoyed a modest reception among 11th/17th-century Ottoman historians due to its resonance with already-established Ottoman historiographical themes; see Cornell Fleischer, "Royal Authority, Dynastic Cyclism, and 'Ibn Khaldûnism' in Ottoman Letters," *Journal of Asian and African Studies*, 18/3-4 (1983), p. 198-220: p. 199. In this context, it should be noted that Ḥāğğī Ḫalīfa (d. 1067/1657) may well have borrowed his dim view of occultism from Ibn Ḫaldūn; see n. 129 below.

84 See Godefroid de Callataÿ, "Who Were the Readers of the *Rasāʾil Ikhwān al-Ṣafāʾ*?" *Micrologus*, 24 (2016), p. 269-302; on the association of Ismaʿilism and lettrism in the Maghrib

9th/15th-century Sunni thinkers such as Yazdī, Ibn Turka and ʿAbd al-Raḥmān al-Bisṭāmī, like Būnian esotericist reading circles of the previous century,[85] pointedly sported the handle Iḫwān al-Ṣafāʾ to announce their cognate occultist project may be regarded as a polemical retort to Ibn al-Qayyim and his ilk in particular; this includes Yazdī's opponent, Zayn al-Dīn Ḫʷāfī (d. 838/1435), a puritanical sufi based in Herat, who denounced the ambitious intellectuals of western Iran as neo-Ismaʿilis (*naw-mulḥidān*, *maḏhab-i malāḥida-yi ǧadīd*).[86]

We turn now to an examination of the main elements of Ibn Ḫaldūn's impressively synthetic anti-occultism argument on theological, natural-scientific, social and political grounds.

In the first place, Ibn Ḫaldūn's position is founded on a basic pessimism with regard to human knowledge. He flatly declares it impossible to attain to a complete knowledge of reality in a way that would validate the claims of disciplines like astrology or metaphysics, regardless of methodology used; certain truths about the cosmos can only be known through revelation, which is only vouchsafed a privileged few.[87] Occultist or philosophical pretensions to comprehensive knowledge, whether rational or superrational, are therefore simply false (and thus harmful to science) and antithetical to religion (and thus harmful to society).[88] Indeed, in this respect, philosophy is more worthless than both astrology and alchemy, since the objectives of the latter two sciences, however harmful and unattainable, are at least theoretically possible, whereas philosophy's objectives are not even theoretically possible.[89] For Ibn Ḫaldūn, as for Ibn al-Qayyim, only knowledge of the religious sciences can protect one from the pitfalls of the rational sciences, including the occult sciences. Most damningly, these sciences tend to induce unwarranted arrogance in their proponents, whose claims to knowledge frequently exceed the scope of the scientific or philosophical methodologies they use, which are thus made to tread on revelation's turf. A proper knowledge of the natural and mathematical sciences should serve only to deepen faith, never to replace it; and the pursuit

see Michael Ebstein, *Mysticism and Philosophy in al-Andalus: Ibn Masarra, Ibn al-ʿArabī and the Ismāʿīlī Tradition*, Leiden, Brill, 2014.

85 Noah Gardiner, *Esotericism in a Manuscript Culture: Aḥmad al-Būnī and His Readers Through the Mamlūk Period*, PhD dissertation, University of Michigan, 2014, p. 156-157.

86 Binbaş, *Intellectual Networks*, p. 249; Gardiner, *Esotericism*, p. 156.

87 Asatrian, "Ibn Khaldūn," p. 91.

88 *Ibid.*, p. 123. In other words, that which contradicts religion (on knowledge-claim grounds) is inimical to social welfare inasmuch as it offers false hope and distracts people from the pursuit of salvation.

89 *Ibid.*, p. 121.

of some sciences should simply be forbidden by state fiat in the interests of societal health.[90]

Ibn Ḫaldūn's theory of human knowledge is also fundamentally elitist, pivoting as it does on the concept of the *fiṭra*, or inborn nature, as that which alone endows man with the power of prophecy—a potential very rarely realized, and then only in combination with the readiness (*istiʿdād*) of the individual so favored. This concept has a venerable pedigree going back to Abū Bakr al-Rāzī (d. 313/925) and was developed by Ibn Sīnā in particular.[91] It is this emphasis on the primacy of the *fiṭra*, however, that renders his attack on occultism somewhat equivocal. Ibn Ḫaldūn's argument here may be reduced to three premises:

1) Human beings can have no knowledge of the unseen (*ġayb*, *ġuyūb*, *muġayyabāt*) or control over nature (*al-ṭabīʿa*)[92] except by means of psychic power (*al-quwwa l-nafsiyya/l-nafsāniyya*), the only mechanism that allows one to transcend the material realm.[93]
2) Psychic power is inborn (*fiṭra*, *ǧibilla*), and can never be acquired or approximated by occultist methodologies or spiritual exercise (*riyāḍa*), only focused.[94]
3) The licitness of miraculous acts and knowledge of the unseen produced by psychic power depends on the degree of passivity or activity of their subject—the more passive the better—and their good or evil effects.[95]

90 Livingston, "Science and the Occult," p. 608. It should be noted that Ibn al-Qayyim's argument to this end parallels al-Ġazālī's in his *al-Munqiḏ min al-ḍalāl*. *Ibid*., p. 607.

91 *Ibid.*, p. 609; Marquet, "Religion, philosophie et magie," p. 148.

92 On this problematic and unstable term see David Edwin Pingree and Syed Nomanul Haq, "Ṭabīʿa," *EI*².

93 As Ibn Ḫaldūn elaborates, "One should realize that all (magic) activity in the world of nature comes from the human soul and the human mind, because the human soul essentially encompasses and governs nature." Ibn Ḫaldūn, *The Muqaddimah*, III, p. 175; cf. *id., Muqaddima*, III, p. 121. In the later addition in this section it is asserted that saints are magically active through faith (*al-kalima l-īmāniyya*) alone, not through psychic power. *Id., Muqaddima*, III, p. 123-124; *id.*, *The Muqaddimah*, III, p. 180. This would seem to contradict the historian's position elsewhere.

94 Partially contradicting this point, in the same interpolated section it is observed that those without innate magical ability may achieve some results through exercise, but acquired magical ability is decidedly inferior to the innate kind. *Id., Muqaddima*, III, p. 124; *id., The Muqaddimah*, III, p. 180.

95 Our historian explains the difference between miracles and sorcery as follows: miracles are produced "by good persons for good purposes and by souls that are entirely devoted to good deeds," while sorcery is practiced "only by evil persons and as a rule is used for evil

These premises, of course, are designed to guarantee a privileged epistemological position for prophetic inspiration (*waḥy*) and prophethood (*nubuwwa*), without which there can be no religion and therefore no order to society or salvation in the hereafter; they also explain the mechanics of prophetic and saintly miracles (*karāmāt*).[96] By the same token, however, they vouch for the reality of both magic (*siḥr*) and divination (*kihāna*), which are to be respectively conceived of as miracle-working and prophecy of an inferior degree.[97] (As Ibn Ḫaldūn asserts: "No thinking person doubts the reality of magic."[98]) It is primarily to the extent that they involve active subjugation of spiritual entities, moreover, that magic and divination can be classified as sorcery, which the Shariʿa categorically forbids. That is to say, Ibn Ḫaldūn maps such occult operations onto a gamut running from licitness to illicitness, with the deciding factor being the activity or passivity of the practitioner. Thus passive receptivity toward divine influxes in the form of miraculous powers and knowledge of the unseen constitutes sanctity (*wilāya*), while active manipulation of the spiritual realm and its denizens to achieve the same is evil. The invocation and subjugation of devils (*šayāṭīn*) is therefore wholly illicit because self-assertive and always pursued for evil ends (*šarr*); letter magic is relatively licit, at least when performed by saints, because passively receptive to supernal influences; and prophetic miracles are wholly licit.[99] In any event, that which is illicit needs not be uncommon; our empiricist historian assures us that he has

actions." *Id.*, *The Muqaddimah*, III, p. 176. Cf. the distinction made by ʿAllāma Ṭabāṭabāʾī in his *al-Mīzān*, Beirut, al-Aʿlamī, 1971-1974, I, p. 240: "If [miraculous acts] occur by way of an 'advance challenge' (*taḥaddī*), as with most of what is related about the prophets, it is termed an inimitable sign (*āya muʿǧiza*); otherwise it is termed a miracle (*karāma*), or (if it involves supplication [to God] (*duʿāʾ*)) an answer to prayer (*istiǧābat daʿwa*). Of the first category, that which involves recourse to a jinn or spirit or suchlike is termed divination (*kihāna*); that which involves invocations (*daʿwa*), astral magic (*ʿazīma*), spells (*ruqya*) or suchlike is termed magic (*siḥr*)."

96 It should be noted that Ibn Ḫaldūn, like Ibn al-Qayyim, is quite sympathetic toward sufism, though only in its earlier, "purer" phase; see *e.g.* his *Šifāʾ al-sāʾil li-tahḏīb al-masāʾil*, Damascus, Dār al-fikr, 1417/1996.

97 Al-Bāqillānī (d. 403/1014) may be considered a precedent here; in his *Kitāb al-Bayān ʿan al-farq bayn al-muʿǧizāt wa-l-karāmāt wa-l-ḥiyal wa-l-kihāna wa-l-siḥr wa-l-nīranǧāt*, for example, the Ashʿari theologian rigorously distinguishes between miracles on the one hand and magic and divination on the other, this while admitting the reality of the latter. Asatrian, "Ibn Khaldūn," p. 74-76.

98 Ibn Ḫaldūn, *Muqaddima*, III, p. 111: *wuǧūd al-siḥr lā mirya fīhi bayn al-ʿuqalāʾ*.

99 Asatrian, "Ibn Khaldūn," p. 94-95, 104.

verified the reality of various unsavory forms of magic through personal experience (*taǧriba*).[100]

The reality—and general illicitness—of magic being undeniable, Ibn Ḫaldūn proceeds to rank workers of magic as follows:

1) those who exercise influence through their own mental focus (*himma*) alone, the purest form of magic;
2) those who do so by using talismans (*ṭilasmāt*) to harness the power of the spheres, the elements or the properties of numbers, a weaker form of magic associated with astrology in particular; and
3) those who exercise influence through the powers of imagination (*al-quwā l-mutaḫayyila*), which is mere prestidigitation (*šaʿwaḏa, šaʿbaḏa*).[101]

The case of lettrism is particularly instructive for understanding Ibn Ḫaldūn's theory of magic and occultism more generally. In the first place, he holds it the least execrable of the occult sciences precisely through its close connection with sufism, the science of *wilāya*, which potentially lends it a hue of moral unimpeachability: saints, unlike sorcerers, are free of base desires and eschew

100 Ibn Ḫaldūn, *Muqaddima*, III, p. 111-113, 123; *id., The Muqaddimah*, III, p. 160-165, 178-179. Here Ibn Ḫaldūn reports himself as a frequent eyewitness to the efficacy of magic. This includes a form of voodoo, as well as a type of ripping magic: in the Maghrib "rippers" (*baʿʿāǧūn*) travel the countryside killing sheep and goats by magically ripping open their bellies, this in order to extort animals from their owners. In the same section the historian repeats the refrain: "The reality of magic is attested by experience (*šahidat la-hu l-taǧriba*)." Ibn al-Qayyim similarly argues against the use of magic on the basis of its harmfulness (*maḍarra*) rather than its impossibility, in this comparable to disbelief (*kufr*) or singing (*ġināʾ*), both of which most certainly exist and that in spades. Ibn al-Qayyim, *Iġāṯat al-lahfān*, p. 261.

101 Ibn Ḫaldūn, *Muqaddima*, III, p. 110; *id.*, *The Muqaddimah*, III, p. 158-159. All three forms of magic are forbidden by the Shariʿa. *Id., Muqaddima*, III, p. 117; *id.*, *The Muqaddimah*, III, p. 169. Our historian here makes the odd and laughably misinformed claim that the *Ġāyat al-ḥakīm* of Maslama b. Qāsim al-Qurṭubī (d. 353/964), aka the *Picatrix*, remains the final word on the subject of magic, and no one has written on it since, while Faḫr al-Dīn al-Rāzī's *al-Sirr al-maktūm* circulates widely in the east. *Id., Muqaddima*, III, p. 109; *id.*, *The Muqaddimah*, III, p. 157. Indeed, Ibn Ḫaldūn is also largely responsible for the persistent misattribution of the *Ġāya* to the Andalusi mathematician-astronomer Maslama l-Maǧrīṭī (d. after 395/1004) in modern scholarship. Godefroid de Callataÿ and Sébastien Moureau, "Again on Maslama Ibn Qāsim al-Qurṭubī, the Ikhwān al-Ṣafāʾ and Ibn Khaldūn: New Evidence from Two Manuscripts of *Rutbat al-ḥakīm*," *Al-Qanṭara*, 37/2 (2016), p. 329-372: p. 330.

intercourse with devils. Moreover, letter magic (*sīmiyāʾ*) is widely known to be effective in its aims (and therefore valid from a natural-scientific perspective), unlike astrology or alchemy:

> The fact that it is possible to be active in the world of nature with the help of the letters and the words composed of them, and that the created things can be influenced in this way, cannot be denied. It is confirmed by continuous tradition on the authority of many (practitioners of letter magic).[102]

As formulated by al-Būnī and Ibn ʿArabī in particular, the sufi subset of letter magic is known as *ʿilm asrār al-ḥurūf/al-asmāʾ*, the science of the secrets of the letters or names, which is held by its exponents to allow "godly souls (*al-nufūs al-rabbāniyya*) to be magically active in the realm of nature by means of the most beautiful Names of God and the divine expressions that emerge from the letters that encompass all the secrets operative in engendered beings."[103] Here Ibn Ḫaldūn draws an unusual distinction between this type of letter magic and talismanic magic (*ṭilasmāt*) in a manner explicable only in terms of the premises identified above.[104] Letter magic, he says, is of two types. The first emphasizes elemental correspondences (*munāsabāt*)—fire, air, water, earth; the second emphasizes numerology, with a focus on magic squares (*aʿdād al-wafq*). In both cases the practitioner is the passive object of divine unveiling (*mukāšafa*) or inspiration, and magical results purely incidental, which renders the science licit. Talismanic magic, on the other hand, which people might assume to be a form of letter magic due to similarity of methodology, is in fact distinct, for it has as its explicit aim the active harnessing of spiritual forces (*quwā rūḥāniyya*) by means of astral and

102 Ibn Ḫaldūn, *The Muqaddimah*, III, p. 174; the passage is copied verbatim from *id., Šifāʾ al-sāʾil*, p. 218.

103 *Id., Muqaddima*, III, p. 120; cf. *id., The Muqaddimah*, III, p. 172. On the development of lettrism and its divorce from sufism in the Persianate east during the 8th-9th/14th-15th centuries see Melvin-Koushki, "The Quest," p. 167-283.

104 Note that where Ibn Ḫaldūn classes talismans and *sīmiyāʾ* as distinct sciences, earlier authors of classifications of the sciences, like Šams al-Dīn Muḥammad Āmulī in his *Nafāʾis al-funūn*, include talismans under the rubric of *sīmiyāʾ*, a natural science. Melvin-Koushki, "The Quest," 213-214. Ibn Ḫaldūn clarifies his position: "At the present time, this science is called *sîmiyâʾ* 'letter magic.' The word was transferred from talismans to this science and used in this conventional meaning in the technical terminology of sufi practitioners of magic. Thus, a general (magical) term came to be used for some particular aspect (of magic)." Ibn Ḫaldūn, *The Muqaddimah*, III, p. 171.

numerological correspondences and the ritual burning of incense attractive to the entity in question, thereby linking high elemental natures (*al-ṭabāʾiʿ al-ʿulwiyya*) with low (*al-ṭabāʾiʿ al-sufliyya*); magical results are here the sole objective. In this intentionality, it is more comparable to alchemy, which has to do with the interaction of body with body (*ǧasad fī ǧasad*), while talismans have to do with the interaction of spirit with body (*rūḥ fī ǧasad*). In other words, practitioners of letter magic rely in the first place on their own psychic power to effect change in nature (and that indirectly), while makers of talismans actively seek to harness celestial entities (*rūḥāniyyat al-aflāk*) by binding them with forms or numerical relations. Letter magic that has recourse to such methods instead of passively depending on intuitive inspiration is tantamount to talismanic magic and hence equally illicit. Properly done, moreover, letter magic is far more strenuous an undertaking than is talismanic magic, for while the latter requires minimal exertion and depends for its efficacy on basic natural-scientific principles (*uṣūl ṭabīʿiyya ʿilmiyya*), the former requires great mental energy and spiritual purity to work.[105]

It is here highly significant that Ibn Ḫaldūn embraces both the fundamental occultist and natural-scientific principle of correspondence (*munāsaba*, *tanāsub*) and the explicitly neoplatonic framework it presupposes, the evolutionary Chain of Being (*silsilat al-mawǧūdāt*)—a surprising concession, given his wholesale rejection of philosophy.[106] Having affirmed the natural-scientific validity of magic, in other words, our historian is left to argue against its practice solely on religious and social grounds—that is, in an Augustinian manner;[107] and even this tack is qualified by a seeming leniency toward the letter magic of the sufis, though only in his earlier treatments of the subject.[108]

105 *Id.*, *Muqaddima*, III, p. 120; *id.*, *The Muqaddimah*, III, p. 172-178; Asatrian, "Ibn Khaldūn," p. 101-102; Gardiner, *Esotericism*, p. 309-317. This section is copied verbatim from Ibn Ḫaldūn, *Šifāʾ al-sāʾil*, p. 218-220.

106 *Id., Muqaddima*, I, p. 152-154; *id.*, *The Muqaddimah*, I, p. 194-195; Marquet, "Religion, philosophie et magie," p. 146-147. He likely borrowed it from al-Fārābī and the Iḫwān al-Ṣafāʾ, although he never mentions the latter directly; more notoriously, the Iḫwān al-Ṣafāʾ would also seem to be Ibn Ḫaldūn's source for the idea that man's form is not fundamentally different from that of an ape. See de Callataÿ, "Who Were the Readers."

107 See Steven P. Marrone, "Magic and the Physical World in Thirteenth-Century Scholasticism," *Early Science and Medicine*, 14 (2009), p. 158-185: p. 161-164.

108 On Ibn Ḫaldūn's later and harsher additions to the section on lettrism in the *Muqaddima* see n. 134 below. Gardiner characterizes lettrism's relationship to sufism in Ibn Ḫaldūn's system as follows: "Ibn Khaldūn places great distance between Sufism proper and lettrism, nesting the latter among the least reputable of the intellectual sciences. While it

Magic aside, Ibn Ḫaldūn's epistemologically pessimistic theory of occultism allows him to be somewhat stricter when it comes to divination, as will be discussed below. To briefly summarize his position for our purposes here, however, he again emphasizes that psychic power or *himma* alone can give knowledge of the unseen, and particularly of future events, with specific divinatory techniques acting as mere crutches to this end. The generous degree to which Ibn Ḫaldūn wishes to entertain the validity of divination is exemplified by his outsize discussion of the divinatory art of *zāʾirǧa*, formally a branch of lettrism.[109] Yet because this technique would seem to violate his theory, being a purely mechanical procedure that produces meaningful results in its own right (and in sound rhyme no less), he is constrained to deny the applicability of the principle of correspondence despite his subscription to the same principle immediately prior:

> We have seen many distinguished people jump at (the opportunity for) supernatural discoveries through (the *zaʾirajah*) by means of operations of this kind. They think that correspondence (in form) between question and answer shows correspondence in actuality. This is not correct, because, as was mentioned before, perception of the [unseen (*ġayb*)] cannot be attained by means of any technique whatever.
>
> This and similar (things) are at first suspected as belonging to the realm of the [unseen], which cannot be known. It is thus obvious that it is from the relations existing among the data that one finds out the unknown from the known. This, however, applies only to events occurring

follows the discussion on magic and talismans, and is thus a subcategory of that topic, Ibn Khaldūn's discussion of lettrism in the *Muqaddimah* is also an extension of his critique of the *aṣḥāb al-tajallī* [*i.e.* the 'modern' sufis, including in the first place Ibn ʿArabī and al-Būnī] [...]. [However, his] discussion of lettrism in the *Muqaddimah* as one of the 'intellectual' sciences seems intended [...] to emphasize its alien-ness to Sufism proper [...]. His primary concern seems to be to demonstrate that the science is incoherent, or at least impenetrable to normal discursive apperception [...]. To whatever extent Ibn Khaldūn is willing to entertain the notion that there may be a genuine, sanctified science of letters, he is unwilling to admit that this science—which is inseparable from *kashf*—can be conveyed in books, or by human teachers for that matter, without degenerating into sorcery [...]. His final judgment on the matter is that the science [...] can only be a form of sorcery, even if it is faciliated through practices which in themselves are permissible, such as *dhikr* and other forms of supererogatory prayer." Gardiner, *Esotericism*, p. 313-317.

109 Among other indicators of his sharp interest in the subject, Ibn Ḫaldūn describes in great detail a *zāʾirǧa* procedure he performed. Ibn Ḫaldūn, *Muqaddima*, III, p. 125-161; *id., The Muqaddimah*, III, p. 199-213.

> in (the world of) existence or in science. Things of the future belong to the [unseen] and cannot be known unless the causes for their happening are known and we have trustworthy information about it.[110]

Ibn Ḫaldūn goes on to declare the fundamental lettrist principle that words correspond to outside reality (*muṭābaqat al-kalām li-mā fī l-ḫāriǧ*) to be categorically impossible where divination is concerned—a rather confusing argumentative about-face, since he seems happy to accept just such a possibility as the primary mechanism of letter and talismanic magic.[111] Despite the historian's overt fascination with and emphatic vindication of the *zāʾirǧa* and related techniques, then, in the end he must dismiss them as being but witty games.[112]

His encompassing if qualified embrace of magic and divination notwithstanding, Ibn Ḫaldūn has nothing but contempt for alchemy (*kīmiyāʾ*) and astrology (*niǧāma*). These two sciences are, in his view, religiously worthless, rationally unconvincing, socially harmful and politically dangerous; he therefore exhorts rulers to stamp their practice out. For its part, alchemy a) doesn't work, b) can't work, c) is a magnet for fraudsters, and d) when it does work it is simply a form of magic, which is illegal.[113] (It must be emphasized that the context for his argument here is the status of 8th/14th-century Mamluk Cairo as scene to a veritable alchemical renaissance, this through the labors of the shadowy but prolific Aydamir al-Ǧildakī, second only to (pseudo-)Ǧābir b. Ḥayyān in his pivotal influence on Arabic alchemy;[114] thus did Aḫlāṭī, phy-

110 *Id., The Muqaddimah*, I, p. 245 (I substitute 'unseen' for Rosenthal's more problematic 'supernatural'); *id., Muqaddima*, I, p. 186.

111 *Ibid.*, I, p. 186.

112 *Id., The Muqaddimah*, III, p. 227.

113 Asatrian, "Ibn Khaldūn," p. 104-109. Having reclassified alchemy as a subset of magic, Ibn Ḫaldūn identifies Ǧābir b. Ḥayyān, prototypical Muslim alchemist, as "the chief sorcerer of Islam" (*kabīr al-saḥara fī hāḏihi l-milla*). Ibn Ḫaldūn, *Muqaddima*, III, p. 109; *id.*, *The Muqaddimah*, III, p. 157. Rosenthal's translation of this passage is somewhat garbled and should be corrected as follows: "[Ǧābir b. Ḥayyān] scrutinized the books of this group and derived [therefrom] the Art (*al-ṣināʿa*) [of alchemy]; he studied its essence and brought it out, and composed a number of works [on the subject]. He discussed at great length both this [science] and the art of letter magic (*sīmiyāʾ*), inasmuch as they are intimately connected; for the transformation of specific bodies from one form into another can only be effected by psychic power (*al-quwwa l-nafsiyya*), not by any practical art (*al-ṣināʿa l-ʿamaliyya*). As such, [alchemy] is a type of magic, as we will mention in the proper place."

114 On this crucial thinker—as yet otherwise unstudied—see Nicholas Harris, "Better Religion through Chemistry: Aydemir al-Jildakī and Alchemy under the Mamluks," PhD dissertation, University of Pennsylvania, forthcoming; and Harris's article in this special issue.

sician-alchemist to Barqūq and ringleader of the Iḫwān, meet great acclaim in the same city.) As for astrology, Ibn Ḫaldūn passes judgment as follows:

> Thus, the worthlessness of astrology from the point of view of the religious law, as well as the weakness of its achievements from the rational point of view, are evident. In addition, astrology does harm to human civilization. It hurts the faith of the common people when an astrological judgment occasionally happens to come true in some unexplainable and unverifiable manner. Ignorant people are taken in by that and suppose that all the other (astrological) judgments must be true, which is not the case. Thus, they are led to attribute things to some (being) other than their Creator. Further, astrology often produces the expectation that signs of crisis will appear in a dynasty. This encourages the enemies and rivals of the dynasty to attack (it) and revolt (against it). We have (personally) observed much of the sort. It is, therefore, necessary that astrology be forbidden to all civilized people, because it may cause harm to religion and dynasty.
>
> The fact that it exists as a natural part of human perceptions and knowledge does not speak against (the need to forbid it). Good and evil exist side by side in the world and cannot be removed. Responsibility comes in connection with the things that cause good and evil. It is (our) duty to try to acquire goodness with the help of the things that cause it, and to avoid the causes of evil and harm. That is what those who realize the corruption and harmfulness of this science must do.[115]

He further sets up a simplistic dichotomy: the more rulers patronize prognosticative sciences like astrology and *ǧafr*, the less they patronize true learning; or, the more occult science, the less real science.[116] It is here telling that Ibn Ḫaldūn pointedly elides the main political application of prognosticative sciences in his own day—that is, as a tool for political legitimation and strategic planning of obvious attraction to dynasts—, in favor of scare tactics: they are

115 Ibn Ḫaldūn, *The Muqaddimah*, III, p. 262-263; see also Asatrian, "Ibn Khaldūn," p. 110-113; George Saliba, "Islamic Astronomy in Context: Attacks on Astrology and the Rise of the *Hay'a* Tradition," *Bulletin of the Royal Institute for Interfaith Studies*, 4/1 (2002), p. 25-46. For his part, Ibn al-Qayyim "considers astrology as dead in his time and its practitioners as simply rehearsing (*taqlīd*) the sayings and errors of the astrologers of the past, without always understanding them." Michot, "Ibn Taymiyya on Astrology," p. 150, n. 14. Similar observations were made even by occult philosophers like Ibn Turka, who dismisses the lettrist tradition of the previous century and a half as being but "toughened, jerked meat" and "stale and timeworn." Ibn Turka, *Kitāb al-Mafāḥiṣ*, MS Majlis 10196 ff. 52a, 53b, 157a; Melvin-Koushki, "The Quest," p. 241.

116 Asatrian, "Ibn Khaldūn," p. 93.

a weapon in the hands of the state's enemies, for the cycling of the heavens suggests that forcible cycling—revolution—is possible on earth. (It may be parenthentically asked, however, why the historian's own theory of the cyclical rise and fall of states and extensive meditations on "signs of crisis" should be exempt from the same criticism!)

These rhetorical maneuvers notwithstanding, Ibn Ḫaldūn's condemnation of astrology and alchemy remains confusingly equivocal from a natural-philosophical perspective. This equivocality is necessitated by his underlying agenda: he wishes to reclassify all the occult sciences under the twin rubrics of magic and divination, with psychic power being the primary standard in all cases; any dependency on physical phenomena or objects such as talismans must render an occultist operation less effective. Thus alchemy, when it works (which it occasionally does), must needs be a type of magic, and astrology a type of divination. The only element of the historian's argument that is unequivocal is his affirmation that, regardless of effectiveness, the pursuit of magic or divination for their own sake is forbidden by the Shariʿa, and this because it often leads to acute social and political problems.

But Ibn Ḫaldūn's killing blows are too glancing; his anti-occultism argument, despite such impressive rhetorical and systematizing gymnastics, did not ultimately prove convincing to most intellectuals and elite patrons of the late 8th/14th and following centuries—precisely due to its reactionary yet equivocal nature. That is to say, I would suggest that the primary reason why the historian's offensive against occultism rhetorically fails is that it concedes too much ground to its opponent. For all his arguments about the futility and harmfulness of the occult sciences, Ibn Ḫaldūn's theory of human knowledge requires that magic and divination be quite real and in some cases legitimate. Moreover, in an apparent bid to have his cake and eat it too, to explain the reality of magic he adopts the overtly neoplatonic Chain of Being theory and the natural-scientific principle of correspondence along that Chain, both concepts espoused by all reasonably educated occultists, thus rendering his argument a suspiciously insider one.[117] (Small wonder that his student al-Maqrīzī became an occult enthusiast, or that the historian himself was able, with more than a little hypocrisy, to palm himself off as a seasoned occultist during his famous Damascus interviews with Tīmūr (r. 771/1370-807/1405) in 803/1401.[118]) Even his disallowing of occultism on social grounds is borrowed from the

117 It should be noted that the historian's most important mentor, Muḥammad b. Ibrāhīm al-Ābilī (d. 757/1356), had studied the occult sciences with the Jewish mathematician Ḫallūf al-Maġīlī and disseminated knowledge of the same. Al-Azmeh, *Ibn Khaldūn*, p. 45, n. 29.

118 Ibn Ḫaldūn, *Taʾrīḫ Ibn Ḫaldūn*, ed. ʿĀdil b. Saʿd, Beirut, Dār al-kutub al-ʿilmiyya, 2010, VII, p. 543-552.

occultists themselves, who for centuries had cautioned against the divulgence of their teachings lest the fabric of society be rent asunder and chaos rein supreme.[119] In so invading enemy territory, as it were, Ibn Ḫaldūn's argument is daring, an instance of calculated risk-taking that simply failed to pay off. As with Ibn al-Qayyim, the fact that he thought it necessary to pursue such a risky and equivocal line of argumentation, and this at such length—roughly *eleven percent* of the *Muqaddima* is devoted to the subject—, testifies eloquently to the ascendancy of occultism in his day.

Ibn Ḫaldūn's Argument Against Geomancy

We saw above that Ibn Ḫaldūn readily admits the reality of divination as an inferior form of prophecy, with various degrees of licitness or illicitness based on the activity or passivity of the diviner. But how does he approach the divinatory art (*ṣināʿa*) of geomancy, whose purpose (as he states) is to extrapolate data about the unseen (*istiḫrāǧ al-ġayb*) and acquire knowledge of the *existentia* (*taʿarruf al-kāʾināt*)?[120]

Ibn Ḫaldūn first divides diviners into two classes: *kāhin*s and *ʿarrāf*s. The first class engages in "possession divination," which is superior to the second because less dependent on the material world; it is instead based on trafficking with devils, whose knowledge is strictly partial and contradictory. The second class engages in "inductive (*taḫakkum*) divination,"[121] inferior in its dependency on physical phenomena rather than spiritual beings. These two classes correspond to the first two classes of mages identified above, the second inferior in their dependency on the physical objects that are talismans. This division, however, has only an epistemological significance, not a social one; soothsayers are soothsayers regardless of the technique they employ. As a technique based on making random marks in the sand or on paper, then, geomancy belongs to the second class.[122] It is also a distinctly non-elite art whose exponents

119 See Melvin-Koushki, "The Quest," p. 199, 201-203.

120 Ibn Ḫaldūn, *Muqaddima*, I, p. 174; *id.*, *The Muqaddimah*, I, p. 226.

121 Fahd, *La divination arabe*, p. 113. Such a distinction is reminiscent of that famously codified in Cicero's (d. 43 BCE) *De divinatione* between natural and artificial divination, but was not as systematically made in the Arabo-Persian occultist tradition. On Francis Bacon's valorization of artificial divination—including in the first place geomancy—over natural see n. 62 above.

122 Asatrian, "Ibn Khaldūn," p. 89-91.

are mere commoners (*qawm min al-ʿāmma*), and unfortunately popular in all civilized regions.[123]

Restating his larger argument, Ibn Ḫaldūn then stresses that the unseen (*al-ġuyūb*) cannot be perceived by means of any art whatsoever, whether rationalist or occultist; it is the sole preserve of those elite individuals (*al-ḫawāṣṣ min al-bašar*) whose inborn natures predispose them to return from the realm of sense perception (*ʿālam al-ḥiss*) to the realm of the spirit (*ʿālam al-rūḥ*).[124] The specifics of a divinatory technique are therefore irrelevant except insofar as they help the diviner to achieve focus. The noblest class of geomancers attempt psychic perception by occupying their senses with study of the combinations of geomantic figures, thereby attaining a state of preparedness; complete freedom from sense perception may then lead to the unveiling (*kašf*) of data on the unseen. In other words, soothsaying, properly done by a gifted individual, can unquestionably give knowledge of the future. Such geomancers are superior to those who content themselves with arbitrary or inductive judgment (*taḥakkum*; *i.e.* the direct analysis of a geomantic tableau) and do not seek to transcend corporeal perception; the latter are limited to mere guesswork.[125] But even the best diviners achieve only a limited form of spiritual perception, not the angelic perception that characterizes prophecy. However successful, it remains a type of soothsaying, and is equivalent to divination by means of bones, water or mirrors.[126]

In terms of its epistemological basis as a discipline, geomancy is driven by arbitrary notions and wishful thinking (*taḥakkum wa-hawā*); that is, unlike astrology (itself epistemologically invalid), which is at least based on natural indications (*dalālāt ṭabīʿiyya*), geomancy is based on merely conventional ones (*dalālāt waḍʿiyya*). As such, it is not subject to rational demonstration (*dalīl*) of any kind.[127] (Its claim to scientific status as a branch of mathematics is here simply ignored.[128])

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123 Ibn Ḫaldūn, *Muqaddima*, I, p. 174, 178; *id., The Muqaddimah*, I, p. 226, 233. This is in contrast to astrology, regarding which Ibn Ḫaldūn states, unaccountably, that it is "unusual in Islam and few people cultivate it." *Id., The Muqaddimah*, III, p. 264.

124 *Id., Muqaddima*, I, p. 178; *id., The Muqaddimah*, I, p. 233-234.

125 See Smith, "The Nature of Islamic Geomancy," p. 30-31.

126 Ibn Ḫaldūn, *Muqaddima*, I, p. 178-179; *id., The Muqaddimah*, I, p. 234.

127 *Id., Muqaddima*, I, p. 175; *id., The Muqaddimah*, I, p. 228. In the first version of this section, Ibn Ḫaldūn further specifies that the conventional indications on which geomancy relies come about by means of *awḍāʿ taḥakkumiyya* (in al-Šaddādī's edition *ḥukmiyya*) and *ahwāʾ ittifāqiyya*, or arbitrary indications and chance impulses. *Id., Muqaddima*, IV, p. 85.

128 Marion Smith observes that Ibn Ḫaldūn's fascination with the numerical properties of the *zāʾirǧa*, by contrast, suggests that he was simply not acquainted with the details of the

For Ibn Ḫaldūn, geomancy, in sum, is simply a form of divination like any other, whereby randomly selected forms are used to occupy the senses so the gifted soul can ascend to the realm of the spirit. Any assertion that geomancy is capable in its own right of giving knowledge of the unseen (*i.e.* by the mere interpretation of geomantic figures and their correspondences) makes it but meaningless babble in theory and practice (*haḏar min al-qawl wa-l-ʿamal*).[129]

But his younger colleague was offended: to so defame geomancy, Yazdī retorts, is to be an *ignorant, weak-minded, backward fanatic*.

Yazdī as Occult-Scientific Historian

Šaraf al-Dīn Yazdī, as noted above, was a card-carrying member of the neopythagorean-neoplatonic-monist Iḫwān al-Ṣafāʾ movement; he thus had an intellectual commitment to advancing the cause of occultism in general and lettrism in particular as occupying an epistemological category superior to that of the rational and religious sciences.[130] He was also, naturally, a mathematician, and even worked for a time at the Samarkand Observatory,[131] which further suggests a predisposition toward the embrace of geomancy as an applied mathematical science.[132] While brief, Yazdī's tract in defense of the divi-

geomantic process, which is equally mathematical. Smith, "The Nature of Islamic Geomancy," p. 32-33.

129 Ibn Ḫaldūn, *Muqaddima*, I, p. 179; *id.*, *The Muqaddimah*, I, p. 234. Ḥāǧǧī Ḫalīfa, who used the *Muqaddima*, is similarly dismissive of geomantic methodology. Ḥāǧǧī Ḫalīfa, *Kašf al-ẓunūn ʿan asāmī al-kutub wa-l-funūn*, Beirut, Dār al-kutub al-ʿilmiyya, 1992, I, p. 912; by the same token, he denies lettrism any scientific status (*ibid.*, VI, p. 94). For his part, Ibn al-Qayyim categorizes geomancy (*al-ḫaṭṭ fī l-arḍ, ṭarq*) as but one of the sciences of the Ǧāhiliyya, extending this term to cover philosophers, astrologers and diviners (*kuhhān*), while acknowledging that it, like its various cognate divination techniques—omen interpretation or bibliomancy (*faʾl*), augury (*zaǧr*), scapulomancy (*katif*), lithomancy (*ḍarb al-ḥaṣā*), physiognomy (*firāsa*), lettrism (*ʿilm al-ḥurūf wa-ḫawāṣṣihā*), etc.—, does indeed produce accurate predictions at times, just not reliably. Ibn al-Qayyim, *Miftāḥ*, p. 1434, 1466.

130 He subscribed to an epistemological hierarchy, given clearest expression by Ibn Turka, which posits the religious sciences as foundational and peripatetic-illuminationist philosophy and then sufism as intermediary, with the ascent culminating in the occult sciences, especially lettrism, the science of *walāya* or human perfection. See Melvin-Koushki, "The Quest," p. 315-324.

131 Binbaş, *Intellectual Networks*, p. 65; see Mahdi Farhani Monfared, "Sharaf al-Dīn ʿAlī Yazdī: Historian and Mathematician," *Iranian Studies*, 41/4 (2008), p. 537-547.

132 Thus the status of ʿAlī Qūščī's (d. 878/1474) student and fellow astronomer Niẓām al-Dīn ʿAbd al-Qādir Rūyānī Lāhīǧī (d. 925/1519) as author of the *Miftāḥ-i Mafātīḥ*, a popular geomantic manual. See Melvin-Koushki, "Persianate Geomancy."

natory science therefore epitomizes the Brethren's project as a whole, whose burgeoning popularity in elite Cairene circles is precisely what provoked Ibn Ḫaldūn's (wildly unsuccessful) attack.

Indeed, the epicenter of such interest among the Mamluk ruling elite appears to have been the court of Sulṭān Barqūq himself, who proved himself to be an enthusiastic patron of the occult sciences, as his personal library attests.[133] Ibn Ḫaldūn was understandably alarmed by the growing influence of Aḫlāṭī's occultist clique over his own royal patron—and so sharpened his attacks on lettrism in later revisions of the *Muqaddima*. While the historian's earlier, slightly less hostile treatments of the science penned before his arrival in Cairo leave its legal status ambiguous, unlike that of astrology, alchemy or philosophy, by 797/1394—*i.e.* at the height of Aḫlāṭī's influence at Barqūq's court—he had inserted a new section flatly declaring lettrism to be mere sorcery and hence illegal.[134] To no avail: Barqūq clearly remained unconvinced.

Unlike his teacher Ibn Turka, Yazdī devoted few works to the occult sciences as such, and achieved his fame primarily as a historian and litterateur; yet his œuvre is permeated with occultist, and especially lettrist and astrological, sensibilities. That lettrism was central to his larger intellectual project is borne out in three works in particular: *Kunh al-murād fī wafq al-aʿdād*, on magic squares; *Ḥulal-i muṭarraz*, the first comprehensive formulation of the *muʿammā* or logograph/riddle genre, which presents it as an essential skill in the lettrist's technical repertoire;[135] and *Nikāt-i kalimāt al-tawḥīd fī l-Qurʾān*, aka *Ḥaqāʾiq al-tahlīl*, a lettrist analysis of the qurʾānic *muqaṭṭaʿāt*, or cryptic sura-initial isolated

133 On Barqūq's library and lettrist interests see Gardiner's forthcoming "The Occultist Encyclopedism of ʿAbd al-Raḥmān al-Bisṭāmī."

134 Gardiner, *Esotericism*, p. 317-319. Specifically, the section in which Ibn Ḫaldūn "unequivocally condemns lettrism as a form of sorcery" does not appear in the earlier version of the *Muqaddima* or in the *Šifāʾ al-sāʾil*, whose own equivocal treatment of lettrism was also imported into the *Muqaddima* in its Cairene recension; this section is found only in copies of the work produced in Cairo, including MS Süleymaniye Damad Ibrahim 863 (copied 797/1394 for Barqūq's library) and MS Süleymaniye Yeni Cami 888 (copied 799/1397 and signed by the author), "into which are inserted slips of paper with revisions of certain parts, including a slightly different—though no less condemnatory—version of the [passage in question]." With Gardiner, I here hypothesize that Ibn Ḫaldūn "significantly sharpened his arguments against al-Būnī and other lettrists during his years in Cairo" as "an attempt to wield his position to stem the rising popularity of 'post-esotericist' lettrism that was spreading among communities of Mamlūk military elites and their close advisors"—including in the first place Aḫlāṭī and his students (*ibid.*, p. 319).

135 From Yazdī's lettrist standpoint, moreover, the appearance of the *muʿammā* genre in itself constitutes the harbinger of a new age. MS Majlis 10196, ff. 374b-375a; MS Majlis 31, ff. 2b-3b; Melvin-Koushki, "The Quest," p. 385, n. 25; Binbaş, *Intellectual Networks*, p. 84-85. On logogriphs, including their association with lettrism, see Gernot Windfuhr, "Riddles and

letters, and the statement of faith *lā ilāha illā Llāh* based on Ibn ʿArabī's *al-Futūḥāt al-makkiyya* and Ibn Turka's *Kitāb al-Mafāḥiṣ*.[136] Here and elsewhere, Yazdī identifies the three basic principles on which the platform of the Iḫwān al-Ṣafāʾ movement rested by the early 9th/15th century:

1) the *coincidentia oppositorum*, or union of opposites;
2) the influence of celestial bodies;[137] and
3) the status of the quranic *muqaṭṭaʿāt* as the keys to existence.[138]

These principles were in turn predicated on the twin principles of secondary causation and correspondence of above to below along the neoplatonic-neopythagorean Chain of Being, erected by "the gradual descent of existence from the One"[139]—precisely the point on which Yazdī will call Ibn Ḫaldūn's bluff on both theological and natural-scientific grounds.

The first of the above three principles has a particularly lettrist salience in this context. As asserted forcefully by Ibn Turka throughout his oeuvre, it is the uncreated, all-creative quranic letter-number that is the supreme *coincidentia oppositorum* (*maǧmaʿ al-aḍdād, muʿtanaq al-aṭrāf*); its resolution and transcendence of all dualities (existence vs nonexistence, existence vs essence) renders it the sole legitimate object of metaphysics, the intellect's only vehicle of return to the One. Significantly, Ibn Turka's perennialist championing of this principle was to be closely paralleled by contemporary and later European Renaissance thinkers like Giovanni Pico della Mirandola (d. 1494), Giordano Bruno (d. 1600)

Chronograms," in *General Introduction to Persian Literature*, ed. J.T.P. de Bruijn, London, I.B. Tauris ("A History of Persian Literature", 1), 2009.

136 See *e.g.* MS Süleymaniye, Ayasofya, 1801 (my thanks to Lisa Alexandrin for procuring me a copy); Binbaş, *Intellectual Networks*, p. 99-101. This *Ḥaqāʾiq al-tahlīl*, in turn, is almost certainly the basis for Ǧalāl al-Dīn Dawānī's (d. 908/1502) *Risāla Tahlīliyya*, a similarly lettrist work. See Melvin-Koushki, "The Quest," p. 256-261. For a list of Yazdī's best-known works see Kamāl al-Dīn Ḥusayn Gāzurgāhī, *Maǧālis al-ʿuššāq*, ed. G.R. Ṭabāṭabāʾī Maǧd, Tehran, Zarrīn, 1376/1997, *maǧlis* 49, p. 234; Muḥammad Mufīd Mustawfī, *Ǧāmiʿ-i Mufīdī*, ed. Īraǧ Afšār, Tehran, Asadī, 1340/1961, III, p. 303.

137 This principle in particular was widely considered a universal law of nature in Europe until Newton promulgated his law of gravitation, itself based on the occult-philosophical theory of love as cosmic force. See Lynn Thorndike, "The True Place of Astrology in the History of Science," *Isis*, 46/3 (1955), p. 273-278; and n. 62 above.

138 Binbaş, *Intellectual Networks*, p. 100-101. For a discussion of these principles see Melvin-Koushki, *The Occult Science of Empire*.

139 Ibn Ḫaldūn, *The Muqaddimah*, III, p. 171; cf. *id., Muqaddima*, III, p. 119: *tanazzul al-wuǧūd ʿan al-wāḥid wa-tartībuhu*.

and Nicholas of Cusa (d. 1464), who coined the Latin term.[140] That is to say, letter-number, as the *maǧmaʿ al-aḍdād*, renders the immaterial material; unites Occult (*bāṭin*) with Manifest (*ẓāhir*), First (*awwal*) with Last (*āḫir*); makes the One many and the many One; marries heaven and earth.[141] The quranic dictum "He is the First and the Last, the Manifest and the Occult" (Kor 57, 3) was accordingly seized upon by the Iḫwān al-Ṣafāʾ as their central motto.

Nor did Yazdī separate theory from practice: all three lettrist-astrological principles drove his writing of history. As a consequence, his works in that field, and especially his celebrated and much-imitated *Ẓafarnāma* (*Book of Conquest*), a history of Tīmūr completed in 839/1436, have a tenor that is expressly and unprecedentedly *occult-scientific*. Most famously, Yazdī was responsible for definitively "timuridizing" the astrological title *ṣāḥib-qirān*, Lord of Conjunction; while it had been increasingly deployed by Seljuq, Ilkhanid and Mamluk dynasts from the 7th/13th century onward, it was Yazdī's casting of Tīmūr in the *Ẓafarnāma* as historically preeminent Lord of Conjunction

140 Melvin-Koushki, "The Quest," p. 336, 359; see *e.g.* David Albertson, *Mathematical Theologies: Nicholas of Cusa and the Legacy of Thierry of Chartres*, Oxford, Oxford University Press ("Oxford studies in historical theology"), 2014; Ernst Cassirer, "Giovanni Pico della Mirandola: A Study in the History of Renaissance Ideas (Part II)," *Journal for the History of Ideas*, 3/3 (1942), p. 319-346; Karen Silvia de León-Jones, *Giordano Bruno and the Kabbalah: Prophets, Magicians, and Rabbis*, New Haven, Yale University Press ("Yale studies in hermeneutics"), 1997, *passim*. Henry Corbin emphasized the centrality of this concept in a number of works. See Henry Corbin, *En islam iranien*, Paris, Gallimard, 1971-1972, p. 84-105, 134-150; *id., Temple and Contemplation*, transl. Philip Sherrard with Liadain Sherrard, London, KPI ("Islamic texts and contexts"), 1986, p. 375-376; *id., Avicenna and the Visionary Recital*, transl. Willard R. Trask, New York, Pantheon ("Bollingen series", 66), 1960, p. 152, 201-203, 215, 376; *id.*, *Spiritual Body and Celestial Earth: From Mazdean Iran to Shīʿite Iran*, Princeton, Princeton University Press ("Bollingen series", 91/2), 1977, p. 71-72. In his *Religion after Religion* (Princeton, Princeton University Press, 1999, p. 67-82), Stephen Wasserstrom offers a wide-ranging discussion of the concept in the History of Religions movement and in the context of poetics, politics and ethics; as he shows, Scholem, Eliade and Corbin were all great advocates of the principle, as was Jung, who preferred the term *complexio oppositorum*. See *e.g.* Carl Gustav Jung, *Mysterium Conjunctionis*, Princeton, Princeton University Press, 1970; and David Henderson, "The Coincidence of Opposites: C.G. Jung's Reception of Nicholas of Cusa," *Studies in Spirituality*, 20 (2010), p. 101-113; and Freud attempted to explain the union of opposites in language and religion in his essay "The Antithetical Sense of Primal Words," in *Collected Papers*, ed. Joan Riviere, New York, Basic Books, 1959, IV, p. 184-191.

141 It should be noted in this connection that in the Arabo-Persian encyclopedic tradition talismans are typically defined as letter-magical devices conjuncting celestial virtues with terrestrial objects. Melvin-Koushki, "Powers of One," p. 148.

that made this claim the core of Timurid—and by extension Indo-Timurid or Mughal—universalist imperial ideology. (Thus Emperor Akbar's claim to be a *ṣāḥib-qirān* superior to Tīmūr based on a comparative analysis of their horoscopes; Akbar's grandson Šāh Ǧahān (r. 1037/1628-1068/1657), more sedately, adopted on his accession the royal title *ṣāḥib-qirān-i ṯānī*, the Second Lord of Conjunction, with Tīmūr being *ṣāḥib-qirān-i awwal*, the First.[142]) But equally crucial is the fact, typically ignored in modern scholarship, that Yazdī proves in the preface through an analysis of Tīmūr's horoscope that the conquerer is a historical manifestation of the *coincidentia oppositorum*—his Ascendant having the saturnine firmness (*ṯabāt*) of Capricorn, which is yet a cardinal or moving (*munqalib*) sign, this combination allowing for sweeping revolution (*inqilāb*) in the world—, a claim that admits of lettrist demonstration. Given its deft and succinct synthesis of the three Iḫwānī principles and invocation of the Iḫwānī motto, I translate Yazdī's key passage in full:

> The manifestation of [the Lord of Conjunction's glorious rule] first dawned from the horizon of conquest and victory, inaugurating an era of abundant joy and celebration in which the tongue of laudation does indite:
>
> What a dream whose interpretation is you!
> What a verse whose exegesis is you!
>
> Yea, the splendor of his auspicious brow shone forth like the Sun. For the instant the Sun enters the heaven that is the throne of sovereignty over the seven climes, the world is illumined: the significance of his auspicious rise like the bright and revealing dawn itself, quickly banishing the night that is the disorder of the world, is thus that the ascent of his fortune is the dawning of a day gladder than the New Year festival itself.
>
> The verification (*taḥqīq*) of this statement is as follows: establishing the bases of rule and erecting the walls of caliphate, the blessed being of that holy eminence—Lord of Conjunction, Last Emperor in History (*āḫir al-zamān*)—stands as the firm and unshakable foundation of the fortune of his glorious house; his horoscope must therefore reflect the extreme firmness of that foundation. Yet the affairs of the world are forever subject to change and revolution (*inqilāb*). According to the manifest wisdom of the dictum "He directs the affair from heaven to earth" (Kor 32, 5), then, which indicates the fact that in the workshop of creation the

142 Moin, *The Millennial Sovereign*, p. 36-60.

embroidering of satin—*i.e.* events in the [sublunary] realm of generation and corruption—is a process that requires its tying to what is above, it is necessary that his august Ascendant occupy a sign of the zodiac that is characterized by a firmness that yet does not negate revolution. Now the only sign to have this [dual] quality is Capricorn: it is firm in its elemental association with earth and its celestial-planetary association with Saturn; at the same time, it is among the cardinal (*munqalib*) signs. [His Ascendant being in Capricorn], in the earthy house of Saturn, therefore simultaneously represents absolute stability and total revolution.

An indication more powerful as to stability and perdurance cannot possibly be imagined. According to verifiers (*ahl-i taḥqīq*),[143] moreover, it is an established principle that the ultimate perfection of a given quality can only be achieved through union with its opposite (*bā ḍidd-i ḫūd muʿāniq*), as any consideration of the nature of the Most Beautiful Names (be they exalted) will make evident: "He is the First and the Last, the Manifest and the Occult; He has knowledge of everything" (Kor 57, 3).

On the basis of these subtle and occult (*ġarīb*) premises, then, it should be clear that when firmness and durability are desired, the most appropriate zodiacal sign for governing events in engendered existence is Capricorn. And indeed, the sweet scent of the verity of this argument may be inhaled from the truth-gardens that bloom with the secrets of the revealed isolated quranic letters (*muqaṭṭaʿāt-i ḥurūf-i munzala-yi qurʾānī*).

[Analysis of Temür's horoscope]

The judgments of the seven stars promise plainly:
they will unlock the world.[144]

That this dual astrological-lettrist argument was central to Yazdī's formulation of Timurid imperial legitimacy is further confirmed by the surviving *Dībāča* to

143 Yazdī and his fellow Brethren frequently refer to themselves as the *ahl-i taḥqīq*, a term with strong Akbarian connotations. Binbaş, *Intellectual Networks*, p. 96-104. I have observed elsewhere that these "verifiers" or "investigators" represent an alternative to Ibn ʿArabī school emphasizing the occultist and prophetic elements of the Doctor Maximus's œuvre, this in contradistinction to the more sufi-centric and better studied Qūnawī-Ǧāmī-Nābulusī line. Melvin-Koushki, "The Quest," p. 418-428.

144 Yazdī, *Ẓafarnāma*, ed. Saʿīd Mīr-Muḥammad Ṣādiq and ʿAbd al-Ḥusayn Nawāʾī, Tehran, Kitābḫāna-yi mūza u markaz-i asnād-i maǧlis-i šūrā-yi islāmī, 1387/2008, I, p. 236-238.

his unfinished *Fatḥnāma-yi ṣāḥib-qirānī*, written in 828/1424, which develops a similar theme.[145]

It is thus evident that, for Yazdī and his fellow Brethren, astrological arguments must always be supported with lettrist ones. But the latter may also be used alone to clinching effect; as Yazdī asserts of Tīmūr elsewhere in the *Ẓafarnāma*:

> Among the most wondrous of signs and gladdening of tidings is the fact that the date of His Majesty's accession rests on the mighty and secure foundation constituted by the uncreated revealed letters that open sura al-Baqara—the centerpiece of the Speech of the All-Knowing King—, [to wit, ALM (= [7]71)].[146] This meaningful synchronicity (*ittifāq-i ḥasana*) must needs inspire great hope in the world and its denizens as to the perdurance and permanence of his felicitated rule—a period of which it may well be said, without exaggeration, that it is superior to all other eras and epochs to the same degree that the Holy Sanctuary [of Mecca] is superior to all other places and regions.[147]

And again:

> The age of that blessed and holy eminence reached 71 years, corresponding to the value of ALM, which opens the greatest of quranic suras; and the length of that matchless lord's rule as sole sovereign was 36 years, corresponding to the value of the three letters that constitute that most excellent of all utterances, to wit, "There is no god but God" (*lā ilāha illā Llāh = LAH*). Indeed, that the fragrant Profession of Oneness (*kalima-yi tawḥīd*) thus seals the sum of the unprecedented words and deeds that issued forth from that oceanic broadcaster of justice is another sure sign of the perfection of his felicitated state. And the fact that the number of offspring from his seed, including both sons and grandsons, is likewise 36, one for each year of his auspicious and glorious rule, is

145 On this work see Binbaş, *Intellectual Networks*, p. 215-216. Specifically, there Tīmūr's claim to being *āḫir al-zamān*—a blatantly messianic title—is proved on lettrist grounds. Yazdī is thus the direct model for Dawānī, ideological mainstay of the Aqquyunlu state, who likewise puts Uzun Ḥasan's (r. 861/1457-882/1478) imperial legitimacy on a primarily lettrist footing in his seminal *Aḫlāq-i ǧalālī* and elsewhere. See Melvin-Koushki, *The Occult Science of Empire*.

146 *I.e.* 1370 CE.

147 Yazdī, *Ẓafarnāma*, I, p. 404-405.

> also productive of hope and justifiable expectation that the sublimity and magnificence of this felicitated and righteous padishah shall not be limited to these his days of physical rule over the world, that abode of deception, but rather perpetuated by his worthy successors.[148]

Such passages cannot be dismissed as mere rhetorical embellishment by the post-Enlightenment occultophobic historian: they epitomize *neopythagorean historiography*. The *Ẓafarnāma*, in other words, that ornate masterpiece, pivots on the proposition that time and space are but *semantic constructs*, to be deconstructed, and reconstructed, by the lettrist historian. The prevailing tendency in scholarship on the development of Timurid ideology to elide lettrism thus does considerable violence to the sources, and renders Yazdī's astrological-lettrist theory of history illegible: for the Timurid historian clearly considers the latter science to be epistemologically superior to all others, including even astrology.

The extent to which this represents a radical departure from both philosophical-scientific and political precedent cannot be overstated; and this departure was first wrought by Ibn Turka, who elevated his brand of philosophical lettrism to the status of sole universal science, encompassing even astronomy-astrology within its aegis. He did so, moreover, similarly in committed service to the Timurid dynasty. A few years after his return from Cairo in *circa* 810/1408—two years after Ibn Ḫaldūn's death in the same city—, Ibn Turka wrote his first lettrist works for his first Timurid patron: Tīmūr's grandson and would-be successor Iskandar Sulṭān (r. 812/1409-817/1414), the ambitious ruler of Fars, whose bid for control of the Timurid Empire was soon stymied by his more conservative uncle Šāhruḫ (r. 811/1409-850/1447).[149]

Nevertheless, and as is well known, Iskandar established a culturally brilliant court in Shiraz and Isfahan, and heavily patronized astronomy-astrology in particular. It is not an accident that one of the only two Islamicate illuminated horoscopes to have reached us, and certainly the most gorgeous, is Iskandar's.[150] What is less well known, however, is the fact that Iskandar seized upon precisely Ibn Turkian lettrism as central prop, with astrology, to his claim

148 *Ibid.*, II, p. 1295.

149 This includes his seminal *Risāla-yi Ḥurūf*, completed in 817/1414; a summary, edition and translation are provided in Melvin-Koushki, "The Quest," p. 161, 463-489.

150 See *e.g.* Anna Caiozzo, "The Horoscope of Iskandar Sultān as a Cosmological Vision in the Islamic World," in *Horoscopes and Public Spheres: Essays on the History of Astrology*, eds Günther Oestmann, H. Darrel Rutkin and Kocku von Stuckrad, Berlin-New York, De Gruyter ("Religion and society", 42), p. 115-144.

to *saint-philosopher-kingship*. He states this quite explicitly in the preface he wrote for a comprehensive Persian manual of mathematical astronomy (*ʿilm-i hayʾat*), *Ǧāmiʿ-i sulṭānī*, possibly his own work as well. In this preface, the Timurid ruler provides an overview of his personal intellectual and spiritual development as proof of his status as perfect man (*insān-i kāmil*) and universal ruler, sultan-caliph of realms both seen and unseen. We are here far beyond the Abbasid import of these terms. The impressive Islamic universalism of Iskandar's self-styling aside, he then claims a total mastery of "all the traditional and rational sciences in both fundamental principles and applications," including theology, philosophy and sufism. But he finally declares astrology (*ʿilm-i nuǧūm*) to be of all sciences the most useful to kings—with the exception of lettrism (*ʿilm-i ḥurūf*), the sole universal science in its status as chiefest branch of the science of divine unity (*ʿilm-i tawḥīd*).[151] Iskandar's *Dībāča* thus constitutes the first formal expression of the peculiarly Timurid occult-scientific imperial platform that would be enshrined in Yazdī's histories, and thence serve as model for all successor imperial states of the Persianate world through at least the 11th/17th century.[152]

It must be emphasized, however, that Ibn Turka and Yazdī were transformed into occultist ideologues of universalist Islamic empire precisely at the Mamluk court. Sulṭān Barqūq, an aspiring *ṣāḥib-qirān*, would seem to have been powerfully attracted to the model of millennial sovereignty being developed by his Timurid courtiers, for all that he declined (like Tīmūr himself) to overtly claim saint-philosopher-kingship in the manner of Iskandar.[153] That is to say: seeking

151 *Dībāča-yi Ǧāmiʿ al-Sulṭānī*, in Šaraf al-Dīn ʿAlī Yazdī, *Munšaʾāt*, ed. Īraǧ Afšār and Muḥammad Riḍā Abūʾī Mahrīzī, Tehran, Ṯurayyā, 1388/2009, p. 207-211. With his neopythagorean proclivities, Iskandar stands cognate to such ancient philosopher-kings as Archytas of Tarentum, one of the three most important pythagorean philosophers and military leader of a powerful Greek city-state in the first half of the 4th century BCE, who is often cited in lettrist works of the Timurid period; see Carl A. Huffman, *Archytas of Tarentum: Pythagorean, Philosopher and Mathematician King*, Cambridge, Cambridge University Press, 2005.

152 See Melvin-Koushki, "Early Modern Islamicate Empire"; Binbaş, *Intellectual Networks*, p. 97, 272-274. Needless to say, the influence of Ibn Turka's lettrist-neopythagorean project was hardly limited to the political sphere. As I argue elsewhere, it provided the primary impetus behind the celebrated *mathematization of astronomy* by the members of the Samarkand Observatory of Uluġ Beg (r. 811/1409-853/1449); it is no accident that its construction was begun the same year (823/1420) Ibn Turka completed his *Kitāb al-Mafāḥiṣ*, which calls precisely for such a mathematizing project. See Melvin-Koushki, "Powers of One"; and n. 42 above.

153 This thesis has been proposed by Noah Gardiner in his paper in development "al-Malik al-Ẓāhir Barqūq as Millennial Sovereign?"; see also Gardiner, "The Occultist Encyclopedism." On Ibn Turka as *timuridizer* of Mamluk literary-imperial culture see my forthcoming

to incarnate the Iḫwānī motto, this Manifest King (*al-malik al-ẓāhir*) became equally Occult.[154]

Such, then, was the emergent neopythagorean-occultist theory of history that Ibn Ḫaldūn found so threatening, and against whose philosophical-scientific bases he railed, however ineffectually, at such length. But perhaps his confused and fideistic polemic was not an overreaction. For the theory put forward by his younger colleague and fellow Cairene Yazdī, being heavily astrological, is predicated precisely on a *cyclical* model of historical change—and was hence in direct competition with his own. Yet the prospect offered to dynasts by Ibn Ḫaldūn's leveling theory, which condemns them to be forever ground under by the cycling of history between two poles, could not but be far less attractive than Yazdī's to the ambitious millennial sovereigns of the post-Mongol era. The former simply moralizes, imprisoning rulers in the binaries of time and the strictures of Shariʿa; it offers in return only the humble rewards that accrue through the strict enforcement of piety on an always erring society. But the latter promises to historically empower and spiritually perfect—to *divinize*—its royal patrons by means of lettrism. Not only can the future be read by the aspiring (occult) philosopher-king: it can also be *rewritten*. Where Ibn Ḫaldūn posits a ceaseless cycling between nomad and settled and attempts to legislate the divorce of occult from manifest, then, Yazdī, the *humanist*, holds up Amīr Tīmūr as *coincidentia oppositorum*, transcendent of all binaries in his historical embodiment of the quranic *muqaṭṭaʿāt* and so an imperial expression of the One.

"Imperial Talismanic Love: Ibn Turka's *Debate of Feast and Fight* (1426) as Philosophical Romance and Lettrist Mirror for Timurid Princes." It should be noted in this connection that Ibn Turka's first major work, completed in Cairo, is an ornate commentary on the *Tāʾiyya l-kubrā* of Ibn al-Fāriḍ (d. 632/1235), that seminal teaching text of the Ibn ʿArabī school; uniquely, his *Šarḥ-i Naẓm al-durar* adapts the new Arabic anthology-as-commentary genre developed by Mamluk litterateurs to the contemporary Persian literary context. See Melvin-Koushki, "The Quest," p. 47, 389-400. Saʿīd al-Dīn al-Farġānī's (d. 700/1301) earlier (and strictly nonliterary) commentary on the same is the only Akbarian text that the poorly-informed Ibn Ḫaldūn is properly familiar with, and hence a primary basis for his rejection of decadent modern sufism. Knysh, *Ibn ʿArabi*, p. 197. That Ibn Turka produced a hybrid Mamluk-Timurid work on precisely the *Tāʾiyya* in the same year (806/1404) that his fellow Cairene Ibn Ḫaldūn finished the final version of the *Muqaddima* therefore suggests it as both a literary and a political act.

154 Ibn Turka makes just this imperial-incarnational imperative explicit in his ornate *Munāẓara-yi bazm u razm*, written in 829/1426 for Bāysunġur b. Šāhruḫ (d. 837/1433), which synthesizes several Mamluk-Timurid Arabo-Persian literary genres to counter Ibn Ḫaldūn's doggedly dualist treatment of the sword *vs* pen trope in particular. So I argue in Melvin-Koushki, "Imperial Talismanic Love."

As a theoretician of history, Yazdī, in sum, is a radical monist; Ibn Ḫaldūn, an old-fashioned dualist. And it would seem only dualists can appreciate the Tunisian historian's anti-transcendence: witness his pointed rejection by a post-Mongol Islamicate historiographical culture prizing the millenarian possibility of universalist-monist imperial transcendence—but his eager reception, some four centuries later, by French colonialists and their orientalist heirs, who crippled the Arab-Berber colonized with Ibn Ḫaldūnian binaries.[155]

Yazdī's Rebuttal

> [S]ome that are perverse [...] by their rash ignorance may take the name of Magick in the worse sense, and [...] cry out that I teach forbidden Arts, sow the seed of Heresies, offend pious ears, and scandalize excellent wits; that I am a sorcerer, and superstitious, and divellish, who indeed am a Magician: to whom I answer, that a Magician doth not amongst learned men signifie a sorcerer, or one that is superstitious, or divellish; but a wise man, a priest, a prophet [...].
>
> HEINRICH CORNELIUS AGRIPPA (D. 1535)[156]

Given such fundamentally different outlooks and mutually exclusive projects, the degree to which Yazdī does in fact agree with the senior historian's analysis of geomancy specifically and occultism generally is perhaps surprising. In his tract in defense of the science, introduced and translated below, Yazdī readily acknowledges that a geomancer does not obtain knowledge of the unseen by means of his art directly, but rather by virtue of innate readiness (*istiʿdād*) and receptivity (*qābiliyyat*) (combined, of course, with thorough study, purity of intention, intensity of concentration and rigorous observation of all the science's conditions)—shades of Ibn Ḫaldūn's doctrine of *fiṭra*. He fully endorses the latter's statement—"Things of the future belong to the [unseen] and cannot be known *unless the causes for their happening are known and we have trustworthy information about it*"[157]—with the proviso that geomancy does in fact provide trustworthy information about the unseen by way of the encompassing occult virtue of uncreated, all-creative number, which alone gives structure to

155 On the inextricability of modern Ibn Ḫaldūn studies from French colonialism in North Africa see *e.g.* Simon, *Ibn Khaldūn*, p. 23-32.

156 "To the Reader," in *Three Books of Occult Philosophy*, transl. John French, London, R.W. for Gregory Moule, 1651.

157 Ibn Ḫaldūn, *The Muqaddimah*, I, p. 245 (emphasis added).

reality. Or rather, he takes this statement further: the geomancer does not access knowledge of the unseen at all, but simply extrapolates forward along the chain of physical events on the basis of reliable data, since number is equally operative at all levels of existence.[158] The future thus belongs not only to the realm of the unseen but potentially to that of the seen. The geomancer, moreover, is a passive agent in this process, which is therefore licit; the effectiveness of a geomantic procedure is entirely based on intuitive perception (*ḏawq*), the faculty targeted by divine unveiling (*kašf*). As such, geomancy is not subject to rational proof (*burhān*), though it is hardly arbitrary as a consequence. Finally, Yazdī concurs with Ibn Ḫaldūn's assessment that the art holds a particular attraction for the unscrupulous, the ignorant and the incompetent, though this unpleasant social fact does not invalidate it; proper training is the answer.

So much for a meeting of historians' minds; Yazdī proceeds to parry Ibn Ḫaldūn's blows (and land a few of his own) with several precise counterstrokes:

First, he calls out the equivocality of the latter's argument by reasserting the principles of correspondence and secondary causation as the natural-scientific basis for divination as for magic, without which the Chain of Being, selectively and indeed mercenarily invoked by Ibn Ḫaldūn, cannot possibly cohere. To deny that the dots and dashes of a geomantic reading reflect the ordered nature of existence in their own right is to deny secondary causality (*i.e.* in the manner of the occasionalist Ašʿarīs), without which principle we cannot hope to understand the world.[159]

158 This argument is in conscious keeping with Kor 27, 65: "None knows the unseen in the heavens and earth except God." Cf. the Brethren of Purity's similar defense of the philosophical and religious legitimacy of judicial astrology: "Know that many people assume that the science of judicial astrology lays claim to knowledge of the unseen (*iddiʿāʾ al-ġayb*), but this is not at all the case. To know the unseen is to know what is or will be without inferring this from any cause (*bi-lā istidlāl wa-lā ʿilal wa-lā sabab min al-asbāb*), which is impossible for any creature—including the astrologer, the soothsayer, the prophet, even the angel; such knowledge is God's alone (glorious and majestic is He)." Iḫwān al-Ṣafāʾ, *Rasāʾil*, I, p. 133. It is similarly reminiscent of that put forward by Muḥammad Salīm al-Wāʿiẓ al-Mawṣilī l-Ḥanafī (*fl.* 1144/1732) several centuries later that lettrism should not even be considered an occult science, since its methods and conditions are so clear and well-established, unlike say algebra, which is far more occult and recondite as a science yet never accounted such. Muḥammad Salīm al-Wāʿiẓ al-Mawṣilī l-Ḥanafī, *al-Kawākib al-durriyya fī l-uṣūl al-ǧafriyya*, MS Ankara, Milli A 309/4, f. 20b; Melvin-Koushki, "The Quest," p. 286.

159 Ibn Ḫaldūn evidently had some sympathy for the Ashʿari position; in particular, he lionizes the school as the middle path, and praises its occasionalist doctrine and definition of prophecy. Al-Azmeh, *Ibn Khaldūn*, p. 106; Asatrian, "Ibn Khaldūn," p. 74, 85. At the same time, however, he happily accepts the principle of secondary causation as a

Second, he turns the latter's argument against occultism on religious grounds on its head by asserting that the impressive accuracy of geomantic readings can only serve to strengthen people's faith, not weaken it, for it is a window onto the wisdom and will of God. As such, the science, properly practiced, cannot possibly be heterodox.

Third, he rejects Ibn Ḫaldūn's classification of divination as an inferior form of prophecy; geomancy as a practice is unrelated to prophecy (just as it is unrelated to philosophy), save for the fact that its general rules were first prophetically revealed, then systematized into a science for use by anyone whose intuition is sound and who follows the correct procedure.

Fourth, he corrects Ibn Ḫaldūn's psychological theory of occultism. Geomancy, Yazdī states, is indeed wholly based on intention (*niyyat*) and mentation/imagination (*pindārī*). For geomancy to be effective, a powerful spiritual connection must first be established between intention and action; one must then be able to mentally establish a link or correspondence (*munāsabatī*) with the all-emanating Source (*mabdaʾ-i fayyāḍ*) of neoplatonic cosmology. A lack of concentration, impure motivations or basic ineptitude cannot but produce invalid readings.

Fifth, he provides a lettrist proof, which he clearly holds to be unassailable, as to the validity of geomancy and its pedigree.[160] That its goal is to "obtain information about unknown circumstances and bring clarity to ambiguous or cryptic situations" is confirmed through its numerological correspondence with the divine name Light (*Nūr*). Given the central role of light in the lettrist metaphysics of Ibn Turka and his circle, itself conceived of as an upgrade to contemporary illuminationist philosophy, Yazdī is here effectively declaring: *Q.E.D.*[161]

Sixth, he defends the traditionalist proof, scornfully dismissed by Ibn Ḫaldūn, which posits the prophet Daniel as the inventor of geomancy.[162] Here again his primary argument in support of his interpretation of the relevant hadith is a lettrist one: geomancy bears a numerological correspondence to the name Daniel, so the historical association necessarily holds.

natural-scientific fact but forbids, on purely religious grounds and with rather shocking hypocrisy, any investigation into the causes of things. Ibn Ḫaldūn, *Muqaddima*, III, p. 23-27; *id., Muqaddimah*, III, p. 34-37; see n. 66 above.

160 As Yazdī asserts in his *Nikāt-i kalimāt al-tawḥīd*, lettrist proofs are among those immediately and instinctively grasped and admit of no doubt (*az yaqīniyyāt-i fiṭrī-st wa dar ān hīč šakk u šubha nīst*). Yazdī, *Nikāt-i kalimāt al-tawḥīd*, MS Ayasofya 1801, f. 38b.

161 See Matthew Melvin-Koushki, "Of Islamic Grammatology: Ibn Turka's Lettrist Metaphysics of Light," *al-ʿUṣūr al-Wusṭā*, 24 (2016), p. 42-113.

162 Ibn Ḫaldūn, *Muqaddima*, I, p. 176; *id.*, *The Muqaddimah*, I, p. 229-231.

Finally, Yazdī disagrees as to the social status of geomancy: far from being merely a low form of soothsaying persuasive only to the untutored common ruck and subject to abuse by charlatans, this applied mathematical science, properly practiced, holds considerable interest for members of the intellectual elite—hence Yazdī's (and al-Ṭūsī's and Ḫafrī's) springing to its defense. The younger historian, of course, was himself no plebian slouch; Yazdī and his teacher and friend Ibn Turka were later accounted the two most prominent intellectuals of Shahrukhid Iran, and their works widely studied from India to Anatolia during the 9th/15th and subsequent centuries.[163]

That committed occultists like Yazdī and his fellow Brethren won such acclaim throughout the Persianate world, this while Ibn Ḫaldūn's arguments seem to have largely fallen on deaf ears, suggests that their project was attuned to the increasingly millenarian zeitgeist in a way that the Tunisian historian's was not. Indeed, we saw above that in his anti-occultism (and anti-philosophy) argument, Ibn Ḫaldūn, making common cause with puritanical Ḥanbalīs like Ibn Taymiyya and Ibn Qayyim al-Ǧawziyya, comes across as a conservative reactionary attempting to force the genie back into the bottle—a quixotic enterprise at best, given the historically abysmal success rate of genie-bottling operations. It is hardly surprising, then, that with the backing of Yazdī and other respected intellectuals geomancy became a prominent feature of the philosophical and political landscape of early modern Islamdom: the art of divination, so crucial to the quest for *divinization* pursued by many Turko-Mongol Perso-Islamic kings and other messianic claimants in the runup to the Islamic millennium, was eagerly patronized at the courts of Emperor Akbar in India, Šāh ʿAbbās in Iran and Sulṭān Süleymān in Anatolia, together with a host of lesser monarchs east and west. It is perhaps a mercy that Ibn Ḫaldūn did not live to see occult and manifest so regally married throughout Islamdom.

Yazdī's Tract in Defense of Geomancy

As noted, Yazdī makes the above points in the compass of a short tract written in defense of geomancy, titled simply *Faṣlī dar bāb-i raml*; I provide a translation of it below. This tract is included in a number of manuscript copies of Yazdī's *munšaʾāt*, thought to be compiled by his brother Qiwām al-Dīn Muḥammad,

163 Dawlatšāh Samarqandī, *Taḏkirat al-šuʿarā*, ed. Edward Granville Browne, Tehran, Asāṭīr, 1382/2003, p. 340; Melvin-Koushki, "The Quest," p. 61.

aka Ṣā'inī (d. 830/1427),[164] which comprise the historian's correspondence, poetry and incidental writings, as well as several items concerning Ibn Turka and his immediate family; it is often featured in hybrid collections containing other epistolary material from the early Timurid period.[165] The fact that it was not put into circulation as a standalone piece, unlike an even shorter tract on finger counting (*angušt-šumārī*),[166] suggests that Yazdī may have planned to use it as a chapter of a larger work that he never completed.[167] The tract is not dated, and the unsystematic nature of the collections in which it is preserved does not suggest a specific timeframe. Because it directly responds to Ibn Ḫaldūn's arguments in the *Muqaddima*, however, which that historian continued working on until his death in 808/1406, we may speculate that Yazdī composed the tract sometime in the 790s/1390s or early 800s/1400s, perhaps even while still in Cairo. The *Munšaʾāt-i Yazdī* was recently edited and published by the lamented Īrağ Afšār; this edition is the basis for my translation below.[168]

• • •

In the name of God, all-merciful, ever merciful.

After offering praise to the Knowing One (*dānā*) Whose pen of splendid power drew the white and red of dawn and sunset—that stage where light and dark meet—upon the throne of manifestation,[169] and prayers for the Leader who made the glad path of guidance—the way of the folk of felicity—clear to his glorious community via the avenue of lawgiving (God bless him and his House as perpetually as he is remembered by those who remember, though his mention be neglected by the heedless), it may be said that all intelligent and insightful persons agree that the art of geomancy (*fann-i raml*), according to the

164 On Qiwām al-Dīn, see Binbaş, *Intellectual Networks*, p. 33-35. His *taḫalluṣ* Ṣā'inī signals his devotion to Ibn Turka, *laqab* Ṣā'in al-Dīn.

165 See Melvin-Koushki, "The Quest," p. 133. While their content is largely the same, Binbaş distinguishes between a *Munšaʾāt-i Yazdī* and a *Munšaʾāt-i Ṣāʾinī*, with the latter containing poetry written for Iskandar b. ʿUmar Šayḫ and Bāysunġur b. Šāhruḫ. Binbaş, *Intellectual Networks*, p. 111.

166 Yazdī, "Risāla dar ʿAqd-i anāmil," in *Munšaʾāt*, p. 91-93.

167 My thanks to Evrim Binbaş for this observation.

168 Yazdī, *Munšaʾāt*, p. 87-91.

169 On such light-dark imagery as an explicit poetical expression of the Ibn Turkian doctrine of the *coincidentia oppositorum* see Melvin-Koushki, "Of Islamic Grammatology," p. 92-93. Ibn Turka himself employs the same in his *Munāẓara-yi bazm u razm*, studied in Melvin-Koushki, "Imperial Talismanic Love."

leading lights among the ancients and the moderns, was and is prized as being among the most noble, sublime and effective of arts.

Some trace the point (*nuqṭa*) of its genealogy back to the circle (*dāʾira*) of prophecy; some see its method of rendering judgments as deriving from the first principles of philosophy (*ummahāt-i uṣūl-i ḥikmat*); and at times the abominably ignorant make so bold as to accuse its exponents of heterodoxy (*sūʾ-i ʿaqīda*) (I take refuge in God!). These positions variously stem from ignorance, hasty judgment, the entrenchment of old and corrupt opinions or fanaticism (*taʿaṣṣub*).

Now all those who rashly deny the excellence of this noble art appear to do so because they hold to the belief that knowledge of the unseen (*ʿilm-i ġayb*) was only vouchsafed the holy [Prophet], knower of things hidden, and it is impossible for others to attain the same. But it is evident that the geomancer does not in any way claim to have knowledge of the unseen. The most that he claims in this regard is that his art is governed by a number of general principles and rules that were initially discovered through revelation and inspiration (*waḥy u ilhām*), then collected and taught from one generation to the next down to the present day, and that it is possible through knowledge of these principles and rules and total concentration (*ṣidq-i andīša*) on the form (*ṣūrat*) the unseen returns (*radd*) to gain information about how specific situations will unfold at specific times and locations.

This is no perversion of belief; to the contrary, it is a clear proof (*dalīl*, *burhān*) that the order of existents (*niẓām-i mawǧūdāt*) in the Chain of Being (*tartīb-i silsila-yi mukawwanāt*) necessarily manifests according to the eternal knowledge and will and wisdom of God. If, hypothetically speaking, it were not so, and events simply happened randomly (*kayfa mā ttafaqa*) or on the basis of temporary, corruptible principles, then one could never accurately deduce from the dashes and dots produced by the geomancer's stylus (*kilk*) the manner of unfolding of past, future and present events. The truth of this statement will be obvious to the intelligent who know the basis on which geomantic readings are derived.

It being evident, then, that a familiarity with this art can but serve to strengthen one's faith and make it unshakeable, for an uninformed person to suppose it to be somehow heterodoxical is sheerest stupidity and ignorance. Likewise, this estimable art gives us to understand the truth of the saying of the master of creation and centerpiece of the necklace of being (the finest blessings and greetings be to him), "Actions are defined by intentions,"[170] while

170 *Innamā l-aʿmāl bi-l-niyyāt* (the Afšār edition incorrectly has *bi-l-bayyināt*); see *e.g.* al-Buḫārī, *Ṣaḥīḥ, Kitāb Badʾ al-waḥy*, bāb 1, nº 1; etc.

his further statement, appropriate to his status as prophecy's Seal, "I have been given the totalizing words,"[171] indicates the universality of its application.

That is to say, this principle is not limited to those actions and acts of worship and obedience mandated by the Shariʿa, for the indications of good and evil and the signs of benefit and injury that appear in a geomantic reading on the basis of the principles of this estimable art do so solely according to the geomancer's intention (*niyyat*). It is for this reason that geomantic readings (*aḥkām*) vary according to the variation of intention: there are accordingly many forms (*ṣūrat*) that when drawn in relation to one kind of intention predict auspiciousness and the realization of one's desires, but when drawn in relation to another kind of intention predict misfortune and the frustration of one's desires.

Thus those adept in this art first evoke (*istiḫrāǧ*) their mind (*ḍamīr*)—that is, their intention—, and only then derive all the figures of a reading. It is essential that a powerful spiritual connection be established between one's intention and one's action so that a change in the former will cause a corresponding change in the latter.

The fact that, as has been made clear, the figures drawn in a geomantic operation wholly depend on the practitioner's intention perfectly exemplifies the truth of the hadith quoted above, as those of sound understanding will unhesitatingly agree. A further reliable proof in support of this statement is the fact that when Zayd, for example, performs a geomantic operation while focusing his intention on ʿAmr, the reading will reflect only ʿAmr's situation, not that of Zayd or of a third party. The principles and rules of this art comprise many more such secrets and details that cannot be gone into here.

> I opened a page of his discourse:
> secreted speech under its every fold.[172]

The correspondence (*tawāfuqī*) between the number of triplicities (*muṯallaṯāt*) that are possible in a geomantic operation and the numerical value of the [divine] name *Nūr* (Light),[173] [*i.e.* 256], constitutes an intuitive indication as

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171 *Ūtītu ǧawāmiʿ al-kalim*; see *e.g.* Muslim, *Ṣaḥīḥ, Kitāb al-Masāǧid wa-mawāḍiʿ al-ṣalāt*, bāb 1, nº 1199; Aḥmad b. Ḥanbal, *Musnad, Musnad Abī Hurayra*, nºs 7521, 8266; etc. This is a favorite hadith of Ibn Turka's as well; see *e.g.* Melvin-Koushki, "The Quest," p. 454, 455, 480.

172 Awḥadī Marāġaʾī, ghazal: *dar ḫarābāt-i ʿāšiqān kūʾī-st / w-andar-ū ḫāna-yi parī-rūʾī-st*.

173 This value is 50 + 6 + 200 = 256; there are four triplicities in a geomantic tableau (a triplicity being two Mothers or two Daughters from the first row and the one Niece they produce), so 16 × 4 × 4 = 256.

to the benefit and goal of the science of geomancy, that being to obtain information about unknown circumstances and bring clarity to ambiguous or cryptic situations by means of specific methods; and perception comes about solely through the overflowing, knowledge-bearing effulgence of the holy name Light. That is, just as light is the cause and means of the manifestation of things, so too does geomancy constitute a means of lifting the obscuring veils from hidden matters. The fact that the number of geomantic figures multiplied by itself[174] as well as the number of these figures multiplied by the number of houses [in a geomantic tableau][175] are both equivalent to the value [of the name *Nūr*] further reinforces this point.

If someone attempts to practice this art but is unable to benefit from it due to the failure of his readings to match reality, this is due to either a lack of knowledge or a defective technique. For the theoretical aspect of this art is complicated and full of secrets that the leading authorities on the subject (God reward their efforts) do not openly discuss in their writings, so the only way to learn them is through long apprenticeship under a skilled teacher. The superficially inquisitive (*bū l-hawas*) are thus unlikely to have much success in this regard.

In practicing this art one must likewise observe many conditions and follow a certain etiquette; a failure to do so on the part of the practitioner will produce invalid readings. [Unfortunately], most practitioners content themselves with the basics of the art as laid out in introductory manuals on the subject and employ it in pursuit of base, worldly ends; their greed and overweening desire to satisfy bodily cravings distracts them from an inquiry into the finer points of this science and from observing its necessary conditions and proper etiquette. There is no doubt but that it is rare for an accurate impression of reality to imprint itself on such minds.

We must therefore apply ourselves to this art with such purity of intention (*ṣafā-yi niyyat*), unity of purpose (*waḥdat-i qaṣd*) and total concentration (*ǧamʿ-i himmat*) that the soul may thereby achieve a type of correspondence (*munāsabatī*) with the all-emanating Source (*mabdaʾ-i fayyāḍ*) that can call forth its effulgence and evoke hidden things to appear as they truly are and as plainly as can be shown. Those who quaff from the flagons of intuition (*kuʾūs-i aḏwāq*) will find their insight confirmed by the fact that the number of dots in all the geomantic figures [*i.e.* 96] is equivalent to the value of the name Daniel (*Dāniyāl*) (God bless and keep him and our own Prophet).[176] From this

174 *I.e.* 16 x 16 = 256. The actual phrase here is *murabbaʿ-i ʿadad-i aškāl-i raml*.

175 Also 16, including that of the Reconciler.

176 *DANYAL* = 96 (4 + 1 + 50 + 10 + 1 + 30).

equivalency based on the method of the secrets of names and numbers it may be understood that the application of this art can only be effective when it occupies the totality of one's attention in singularity of purpose, all centered on the operation to the extent possible, for this science is entirely predicated on mentation and imagination (*pindārī*).

Among those whom the present writer has seen exemplifying this ideal with true intent through the rigorous pursuit and attainment of total competence in this art is the esteemed master Mawlānā Šams al-Dīn Muḥammad Qaṣṣār Yazdī (God increase his learning!). About this dear man it may be truthfully said that his desire to attain such perfection impelled him to take purpose by the collar, shake out the sleeves of preoccupation with any other activity, tuck up the skirt of endeavor and effort in the belt of singlemindedness and detach himself from all obligations and hindrances for a considerable length of time, and then wholeheartedly enter the path of discipleship; he has thus spent a large part of his life apprenticing himself to teachers, frequenting practitioners and collecting and reading books on the subject. In fact, [he attained to such a level of expertise that] he himself is responsible for establishing many of the principles and rules of this art and for recording rare and useful observations on its practice, and through long habituation he is able to effectively evoke his mind at the beginning of each operation and thereby deliver accurate readings, as geomancers may be expected to do.

My purpose in compiling these words and noting these examples is to encourage the intelligent, when confronted by the inaccuracy of the geomantic readings done by certain false and insincere practitioners and the vacuousness of their predictions, to simply attribute this to their lazy incompetence, and not think of discounting this noble art whose principles and rules were established through prophetic revelation (*waḥy*) and the inspiration (*kašf, ilhām*) vouchsafed purified saints. Indeed, the surest testament to the validity of this science is the statement of the holy Messenger and Seal himself (the highest blessings upon him and all his brethren from among the prophets): "There was a prophet [who practiced divination by] drawing lines, so one may draw the way he did."[177] For this reason some exegetes hold that the term "science" (*ʿilm*)

177 *Kāna nabiyy*[un] *mina l-anbiyāʾi yaḫuṭṭu fa-man wāfaqa ḫaṭṭahu fa-ḏāk*. This prophet is usually taken to be Daniel or Idrīs; see *e.g.* Muslim, *Ṣaḥīḥ*, *Kitāb al-Masāǧid wa-mawāḍiʿ al-ṣalāt*, bāb 8, nº 1227; Abū Dāwūd, *Sunan*, *Kitāb al-Ṣalāt*, bāb 173, nº 931; etc. Some hadith commentators acknowledge this as a prooftext for the licitness of geomantic practice, but a number, including al-Nawawī, al-Ḫaṭṭābī and al-Qāḍī ʿIyāḍ, argue for its illicitness during the Islamic dispensation; see *e.g.* Muḥammad al-Amīn al-Urmī, *al-Kawākib al-wahhāǧ wa-l-rawḍ al-bahhāǧ fī šarḥ Ṣaḥīḥ Muslim b. al-Ḥaǧǧāǧ*, Jeddah, Dār al-minhāǧ, 2009, VIII,

in the verse "Bring me a Book before this, or some trace of a science (*aṯāratin min ʿilmin*), if you speak truly" (Kor 46, 4) refers to the art of geomancy.[178]

Therefore, if a student undertakes a serious study of this science, fully applying himself, yet finds the results disappointing, he is to attribute it to his own lack of receptivity (*qābiliyyat*) and poor aptitude (*istiʿdād*), not to a presumed invalidity of its principles and rules. For every type of work demands a type of aptitude peculiar to it, and this is especially true in the case of this difficult work; in addition to a precise understanding of its principles and procedures, observation of appropriate times and adherence to its rules of etiquette, it requires powerful intuition (*ḥads*) and perfect discernment (*faṭānat*) [on the part of the practitioner]. "Every type of work has its practitioners," [as they say], and "each person finds it easy to do what he was created to do."[179]

Were I to adduce further premises than the ones I already have they would not amount to a formal proof (*burhān*), and there would seem to be no way to categorically establish the validity of these premises in any case. But those possessed of sound intuition (*ḏawq*) and innate talent will recognize them to be free of empty claims and exaggeration and untroubled by either overzealousness or skepticism.

> Anyone who knows her alley
> will know where my goods are from.[180]

God bless and keep the best of His creation, Muḥammad, and his pure and holy House most abundantly; praise be to God, Lord of all the worlds.

Written by the poor wretch ʿAlī l-Yazdī.

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p. 126-128. Ibn Rušd al-Ǧadd (d. 520/1126) likewise judges geomancy to be illicit in his fatwa on the subject, attempting to demonstrate that this hadith in fact represents not an endorsement but a definitive prohibition. Ibn Rušd, *Fatāwā Ibn Rušd*, ed. al-Muḫtār b. al-Ṭāhir al-Talīlī, Beirut, Dār al-ġarb al-islāmī, 1407/1987, I, p. 249-261.

178 See *e.g.* Muḥammad al-Qurṭubī (671/1272), *al-Ǧāmiʿ li-aḥkām al-Qurʾān*, Tehran, Nāṣir-i Ḫusraw, 1364/1985, XVI, p. 179-183; the exegete here argues for the licitness of oneiromancy and omen interpretation or bibliomancy (*faʾl*), but the illicitness of other forms of divination, including ornithomancy (*ṭīra*) and augury (*zaǧr*) (and presumably geomancy).

179 See *e.g.* al-Buḫārī, *Ṣaḥīḥ*, *Kitāb al-Tafsīr: Sūrat 'Wa-l-layli iḏā yaġšā*,' bāb 8, nº 5000; Aḥmad b. Ḥanbal, *Musnad*, *Musnad al-ʿašara al-mubaššarīn bi-l-ǧanna wa-ġayrihim*, *Min musnad ʿAlī b. Abī Ṭālib*, nº 631; etc.

180 Cf. Šāh Niʿmat Allāh Walī, ghazal: *yārī ki zi malak āšnāʾī-st / dānad ki qumāš-i mā kuǧāʾī-st*.